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Research paper

Power-to-X Economy: Green e-hydrogen, e-fuels, e-chemicals, and e-materials opportunities in Africa

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ABSTRACT

Africa has enormous potential to produce low-cost e-fuels, e-chemicals and e-materials required for complete defossilisation using its abundant renewable resources, widely distributed across the continent. This research builds on techno-economic investigations using the LUT Energy System Transition Model and related tools to assess the power-to-X potential in Africa, for meeting the local demand and exploring the export potential of power-to-products applications. In this context, we analysed the economic viability of exporting green e-fuel, e-chemicals and e-materials from Africa to Europe. We also present the core elements of the Power-to-X Economy, i.e., renewable electricity and hydrogen. The results show that hydrogen will likely not be traded simply due to high transport costs. However, there is an opportunity for African countries to export e-ammonia, e-methanol, e-kerosene jet fuel, e-methane, e-steel products, and e-plastic to Europe at low cost. The results show that Africa's low-cost power-to-X products backed by low-cost renewable electricity, mainly supplied by solar photovoltaics, is the basis for Africa's vibrant export business opportunities. Therefore, the Power-to-X Economy could more appropriately be called a Solar-to-X Economy for Africa. The Power-to-X Economy will foster socio-economic growth in the region, including new industrial opportunities, new investment portfolios, boost income and stimulate local technical know-how, thereby delivering a people-driven energy economy. Research on the topic in Africa is limited and at a nascent stage. Thus, more studies are required in future to guide investment decisions and cater to policy decisions in achieving carbon neutrality with e-fuels, e-chemicals, and e-materials.

1. Introduction

Transitioning from fossil fuels such as coal, oil and natural methane to renewable energy (RE) sources across energy-related sectors is made possible by the continued technological innovations and a societal push toward sustainable development (Cherp et al., 2018; Delucchi and Jacobson, 2011; Jacobson and Delucchi, 2011; IEA, 2019a; Lund et al., 2017; Sørensen, 1981). The transition is spurred by worsening climatic conditions, the revisions on the remaining carbon budget, the ongoing global pandemic, the current energy crisis and the skyrocketed price of commodities (IPCC, 2022; IEA, 2022; Breyer et al., 2023a). The core objective of the energy transition is to reduce greenhouse gas (GHG) emissions in energy-related sectors through various forms of defossilisation strategies (IPCC, 2022) in line with the Paris Agreement on climate change, to keep the global temperature below 2°C and pursue efforts to limit the increase to 1.5°C above pre-industrial levels (UNFCCC, 2015). For the case of Africa, it is increasingly concluded that covering all energy needs for a fast-growing demand may be only possible with an energy system dominantly based on renewables (Oyewo et al., 2023). Sterl et al. (2024) emphasise the need for thorough investigation into unique elements of energy transitions research in Africa to inform decisions through clear, objective country-specific analyses.

Reaching net zero GHG emissions and becoming climate neutral is increasingly being discussed by nations (UNFCCC, 2015). Pursuing this path would require massive electrification of end-use via direct and indirect electrification approaches (Lund et al., 2017; Oyewo et al., 2023; Ueckerdt et al., 2021; Breyer et al., 2022; IRENA, 2022a). While direct use of renewable electricity is the most cost-effective, efficient solution and sustainable form of energy utilisation, deep defossilisation of energy-intensive sectors hinges on the indirect electrification approach, i.e., **power-to-X solutions (**IRENA, 2022a; Ram et al., 2020; Lopez et al., 2023a; Galimova et al., 2023a; Lester et al., 2020; Ridjan et al., 2016). For example, there is a need for drastic changes in the production of materials in the **heavy industrial sectors** (Diesing et al.,), such as cement, aluminium, iron and steel, chemicals, and

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petrochemicals (IRENA, 2022a; Ram et al., 2020; Lopez et al., 2023a; Galimova et al., 2023a). **Heavy-duty transport sectors** such as shipping, trucking, and aviation also require alternatives to heavy reliance on fossil fuels (IRENA, 2022a; Brynolf et al., 2022; Kany et al., 2022). These sectors, i.e., industry and transport, are often referred to as hard-to-abate sectors, as their complete defossilisation is infeasible solely by direct electrification (IRENA, 2022a; Galimova et al., 2023a; Bogdanov et al., 2021a, 2021b; Neumann et al., 2023). In hard-to-abate sectors, indirect electrification is an important solution that will contribute to further emission reductions while avoiding the sustainability problems associated with biofuels (Ueckerdt et al., 2021; IRENA, 2022a; Ram et al., 2020; Lopez et al., 2023a; Galimova et al., 2023a; Brynolf et al., 2022; Bogdanov et al., 2021a; ETC, 2018; Lopez et al., 2023b; Merfort et al., 2023). For these reasons, indirect electrification approaches, including **power-to-fuel** (hydrogen, kerosene jet fuel, ammonia, methanol, methane) (Ueckerdt et al., 2021; Galimova et al., 2023a; Palys and Daoutidis, 2022; Breyer et al., 2024), **power-to-chemical** (ammonia, methanol) (Galimova et al., 2023a; Lopez et al., 2023b; Fasihi et al., 2021; Dieterich et al., 2020), and **power-to-material** (plastic, steel, aluminium, carbon fibres) (Lopez et al., 2023a; Keiner et al., 2023a; Otto et al., 2017; Mühlbauer et al., 2024) processes based on **renewable electricity** are vital to meeting these the hard-to-abate demands. **Renewable electricity and hydrogen are the backbone of the Power-to-X Economy** (Ueckerdt et al., 2021; Breyer et al., 2024; Fasihi and Breyer, 2020; Breyer et al., 2023a). Hydrogen is an important precursor in the power-to-X processes, and its relevance is driven significantly by the availability of low-cost renewable electricity. Most power-to-X processes begin with the production of e-hydrogen by electrolysers (Ueckerdt et al., 2021; Breyer et al., 2024; Ball and Wietschel, 2009; Andrews and Shabani, 2012; Parra et al., 2019) that is further synthesised into other e-fuels (Galimova et al., 2023a; Lopez et al., 2023b), e-chemicals (Galimova et al., 2023a; Lopez et al., 2023b; Dieterich et al., 2020), e-plastics (Lopez et al., 2023b; Palm et al., 2016), and e-steel (Lopez et al., 2023a; Otto et al., 2017; Lopez et al., 2022; Devlin and Yang, 2022; Bailera et al., 2021).

Several studies have explored the techno-economic potential of green e-fuels, e-chemicals, and e-materials at different production sites worldwide, in Europe and Africa (Lopez et al., 2023a; Galimova et al., 2023a; Keiner et al., 2023a; van der Zwaan et al., 2021; Timmerberg and Kaltschmitt, 2019; Hampp et al., 2023; ElSayed et al., 2023; Fasihi et al., 2017). As costs fall, **Africa could become a potential hotspot for the low-cost production of e-fuels, e-chemicals, and e-materials required for full defossilisation, using its abundant renewable resources** widely distributed across the continent (Oyewo et al., 2023; Ueckerdt et al., 2021; Lopez et al., 2023a; Galimova et al., 2023a). The low cost of electricity in Africa can be attributed to low-cost solar electricity generation; therefore, the Power-to-X Economy (Breyer et al., 2023a) could more appropriately be called a **Solar-to-X Economy** for Africa (Lopez et al., 2023a; Galimova et al., 2023a). **Africa could emerge as an important exporter of e-fuels, e-chemicals, and e-materials.** As the need for sustainable fuels grows, importing e-fuels from countries with cheap and excellent renewable resource potential is a promising way to contribute and support investments in renewables and power-to-X value chains in the Global South (Lopez et al., 2023a; Galimova et al., 2023a). **Africa, with cheap and excellent renewable resources, could support a global defossilisation strategy**, particularly European countries seeking to diversify their energy import sources while fulfilling their climate targets (Oyewo et al., 2023). In addition to contributing to international climate change mitigation efforts, this study represents an essential step in the research for a 100% RE transition as **power-to-X solutions will drive an integrated energy system development** (Breyer et al., 2022).

The potential of Africa to export energy carriers to high energy demand centres in Europe has been explored by several studies. Sens et al. (2022a). tried to identify the most attractive regions to supply green e-hydrogen to Germany via pipelines. Northern Africa was found to have some of the lowest production costs and also the highest technical supply potential in both 2030 and 2050. However, due to long distance and high transportation costs, the final import costs to Germany from North Africa are close to the import costs from the North Sea (Sens et al., 2022a). In another study, compressed gaseous hydrogen imports from Tunisia to Germany are found to be slightly cheaper than centralised hydrogen production based on offshore wind power in Germany or imports from Argentina (Sens et al., 2022b). Imports of hydrogen, methane, ammonia, methanol, and electricity-based Fischer-Tropsch liquids (e-FTL) fuels to Germany from seven regions across the world, including Egypt and Morocco, have been compared by Hampp et al (Hampp et al., 2023). Imports from Egypt and Morocco have been found to be economically attractive compared to domestic production across different fuels, although for some energy carriers, such as hydrogen or methane via pipelines, Denmark may be more attractive as a supplier due to its close proximity (Hampp et al., 2023). Pfennig et al. (2023). also found many locations closer to the shore with competitive hydrogen prices, among which are Western Sahara and Mauritania. Import costs for e-FTL, e-methanol, e-methane, and e-ammonia to Germany from Morocco have been found to be attractive. Other regions, however, can offer even more attractive final import costs, for example Argentina or Chile (Pfennig et al., 2023). Moritz et al. (2023). estimated supply costs of hydrogen and hydrogen-based energy commodities from 113 countries to Germany via pipelines and shipping, covering many countries in Africa (Moritz et al., 2023). Other studies have considered attractiveness of fuel supply from Namibia and Egypt (Runge et al., 2023), supply costs of hydrogen to Germany from 94 countries across six continents, including many countries in Africa (Brändle et al., 2021), or efficiencies and costs of five different energy carriers from Morocco/Western Sahara to Europe (Hank et al., 2020).

While several studies analysed import options from many regions of the world, most of other studies with inclusion of Africa have focused only on Northern Africa, predominantly on Morocco as a potential renewable fuel supplier. Sub-Saharan African countries were mostly included only in studies with a large set of potential exporter regions, indicating a significant research gap in research on e-fuel and e-chemical trading studies.

The **power-to-X solutions for climate neutrality** is gaining significant attention in policy targets and investments (Palys and Daoutidis, 2022; Breyer et al., 2024; Parra et al., 2019). Furthermore, in response to the current energy crisis, especially in Europe. In this context, this research builds on techno-economic investigations using the LUT Energy System Transition Model (**LUT-ESTM**) (Bogdanov et al., 2021a, 2021b) and related tools to assess the power-to-X potential in Africa, including meeting the local demand and exploring export potential of renewable electricity-based carbon-neutral fuels, chemicals, and materials. In so doing, **we analysed the economic viability of exporting green e-fuel from Egypt to Europe, as well as exporting e-plastic and e-steel from Morocco**. **We also present the core elements of the Power-to-X Economy, i.e., renewable electricity and hydrogen**. An absolute CO_2 emission pathway was examined without negative emission technologies. This implies a phase-out of fossil fuels and carbon offsets through CO_2 removal via direct air capture (DAC), also referred to as direct air carbon capture and utilisation (DACCU) for the production of carbon-neutral e-fuels and e-chemicals.

The paper is structured as follows: **Section 2** describes the methods and the energy system model (ESM) used, **Section 3** presents the results, and **Section 4** interprets the results and discusses their implications. The limitations of the study are presented in **Section 5** . Finally, **Section 6** presents the main conclusions and policy implications.

2. Methods

This section gives an overview of the methods and the mainly used model **LUT-ESTM**. This study assumes the production of green fuels, chemicals, and materials based on renewable electricity. Details on methods, supply chain and transportation of e-fuels, e-chemicals, and e-materials are provided in (Lopez et al., 2023a; Galimova et al., 2023a; Fasihi and Breyer, 2020; ElSayed et al., 2023). Techno-economic assumptions used for e-fuels, e-chemicals, and e-materials production, RE resources potential, regional-specific data, and the definitions of the scenario investigated are presented in this section.

2.1. Description of LUT Energy System Transition Model

LUT-ESTM is a multi-technology, multi-sector, and multi-scenario linear optimisation tool capable of handling hourly sequential temporal resolution for an entire year (Bogdanov et al., 2021a, 2021b). The hourly feature of the model is critical when modelling an energy system with high shares of variable renewable energy to ensure flexible and stable grid operation. The primary objective function of the model is to minimise the total annualised cost of the system as presented in Eq. (1) and comprises all hours of a year using the abbreviations: sub-regions (*r, reg*), generation, storage and transmission technologies (*t, tech*), capital expenditures for technology *t* ($CAPEX_t$), capital recovery factor for technology *t* (crf_t), fixed operational expenditures for technology *t* ($OPEXfix_t$), variable operational expenditures technology *t* ($OPEXvar_t$), installed capacity in the region *r* of technology *t* ($instCap_{t,r}$), annual generation by technology *t* in region *r* ($E_{gen,t,r}$), cost of ramping of technology *t* ($rampCost_t$), and sum of power ramping values during the year for the technology *t* in the region *r* ($totRamp_{t,r}$). The model can analyse the transition pathways in 5-year intervals from now until 2050.

$$\min\left(\sum_{r=1}^{reg}\sum_{t=1}^{tech}(CAPEX_t\cdot crf_t+OPEXfix_t)\cdot instCap_{t,r}+OPEX\mathrm{var}_t\cdot E_{gen,t,r}+rampCost_t\cdot totRamp_{t,r}\right) \tag{1}$$

LUT-ESTM (Bogdanov et al., 2021a, 2021b) incorporates electricity generation, heat generation, energy storage, seawater desalination, transport, transmission, and power-to-X technologies. The **power-to-X technologies** include all fuel conversion and sector bridging technologies, such as electrolysers, heat pumps, Fischer-Tropsch liquid (FTL) plants, DAC units, methanation, ammonia synthesis, and methanol synthesis, to mention a few. Power prosumers and individual heating systems are also considered in the energy system. Exogenously, individual heating and power consumers are optimised in hourly resolution. Prosumers' heating and electricity consumption is optimised exogenously by hourly resolution. The prosumer's objective function encompasses the costs associated with electricity generation and storage as well as heating systems, electricity consumed by the grid, and fuel consumed by boilers, income via electricity feed-in to the distribution grid is deducted from the total annual cost. The target function for the prosumers is presented in Eq. (2) and comprises all hours of a year using the abbreviations: generation and storage technologies (*t, tech*), capital expenditures for technology *t* ($CAPEX_t$), capital recovery factor for technology *t* (crf_t), fixed operational expenditures for technology *t* ($OPEXfix_t$), variable operational expenditures technology *t* ($OPEXvar_t$), installed capacity of technology *t* ($instCap_t$), annual generation by technology *t* ($E_{gen,t}$), retail price of electricity (*elCost*), feed-in price of electricity (*elFeedIn*), annual amount of electricity required from the grid (E_{grid}), annual amount of electricity fed-in to the grid (E_{curt}).

$$\min\left(\sum_{t=1}^{tech}(CAPEX_t\cdot crf_t+OPEXfix_t)\cdot instCap_t+OPEX\mathrm{var}_t\cdot E_{gen,t}+elCost\cdot E_{grid}+elFeedIn\cdot E_{curt}\right) \tag{2}$$

In determining a cost-optimal path for the energy transition, technical and financial assumptions for various technologies are based on market developments and findings from the scientific literature. In all scenarios studied, the weighted average cost of capital (WACC) ranged from a maximum of 12% in 2020 to 11.7%, 11%, 9.8%, 8.5%, 7.5%, and 7% in subsequent time steps. The cost assumptions for all technologies, including generation, storage, transmission, and power-to-X, are shown in Tables 1–3.

The model incorporates a 4% annual growth rate limit on RE capacity share to avoid system disruption. No new fossil and nuclear technologies are allowed, except gas turbines since natural gas can be substituted with e-fuels during the transition. These constraints apply to a fully RE system and are subject to changes depending on the scenario. Additional information on the model and mathematical functions can be found in Oyewo et al. (2022)). Fig. 1 illustrates the schematic description of LUT-ESTM highlighting key components.

2.2. Description of power-to-X systems

This section presents the key technologies and processes for producing e-fuels, e-chemicals, and e-materials. The fuels produced include e-hydrogen, e-FTL, mainly e-kerosene for jet fuel and e-naphtha for the chemical industry, as well as the chemicals e-ammonia and e-methanol for the chemical industry and marine fuels. In addition, e-methanol can be further converted to e-kerosene, e-gasoline, and e-diesel (Olah et al., 2009; Ruokonen et al., 2021). e-Hydrogen is the main precursor in many power-to-X processes, i.e., power-to-hydrogen-to-X fuels, feedstocks, and materials. The key component of the e-hydrogen production chain includes renewable electricity supply, seawater reverse osmosis desalination, and electrolysers. These components also form the basis for the other investigated e-fuels, e-chemicals, and e-materials. Additional components are required to produce e-FTL, including CO_2 DAC units, reverse water-gas shift plants, FTL plants, and product upgrading units based on Fasihi et al. (2017, 2016). The production of green electricity-based methanol (e-methanol) also uses CO_2 DAC units in addition to single-unit methanol synthesis plants based on Fasihi and Breyer (2024). The production process of green electricity-based ammonia (e-ammonia), i.e., power-to-ammonia, requires additional components, including air separation units (ASU) to capture nitrogen from air that is coupled to Haber-Bosch ammonia synthesis plants based on Fasihi et al. (2021). The production chain of e-materials, including steel and plastic, is based on Lopez et al. (2023b, 2022). The e-fuel, e-chemicals, and e-materials value chains are depicted in Fig. 2. The techno-economic assumptions used for the power-to-hydrogen-to-X fuels, feedstocks and materials are summarised in Table 1. The technical and financial assumptions are made for various technologies based on market developments and findings from the scientific literature. The estimation of the demand for e-fuels, e-chemicals and e-materials production for exports from Africa, using Egypt and Morocco as a representation case study, is based on ElSayed et al. (2023)., Galimova et al. (2023a, 2023b, 2023c)., and Lopez et al. (2023a, 2023b).

The CO_2 DAC units applied in this study are low-temperature solid sorbent (LT DAC) systems. The LT DAC systems are favourable due to lower heat supply costs and the possibility of using waste heat from other systems. The LT solid sorbent DAC system specification is based on Fasihi et al. (2019). Each unit contains a solid sorbent that captures and releases CO_2 via temperature swing adsorption. Electricity and heat at 100°C are required to capture and regenerate CO_2. The LT DAC configuration is modified based on ElSayed et al. (2023). The levelised cost of CO_2 DAC is estimated based on Breyer et al. (2019, 2020a). The structure of the CO_2 supply by DAC units is presented in Fig. 3. The techno-economic assumption of CO_2 DAC, transportation, and storage is summarised in Table 2 and steelmaking in Table 3.

2.3. Regional-specific information

The power, heat, transport, and desalination sectors in Africa have been modelled in detailed full sector coupling and multi-nodal approach described in (Oyewo et al., 2022). For a comprehensive analysis,

Table 1
Power-to-X sector key specifications. Abbreviations: Capital expenditure, Capex, fixed operational expenditure, Opex fix, variable operational expenditure, Opex var.

Technology	Parameter	Unit	2020	2025	2030	2035	2040	2045	2050	Ref.
Alkaline water electrolyser	Capex	€/$kW_{el,LHV}$	563	411	313	267	243	219	204	Breyer et al., (2015); IRENA (2020a)
	Capex	€/$kW_{H2,LHV}$	803	586	446	381	347	313	291	
	Opex fix	€/$kW_{H2,LHV}$·a	28.1	20.5	15.6	13.3	12.1	11.0	10.2	
	Opex var	€/$kWh_{H2,LHV}$	0.0014	0.0014	0.0014	0.0014	0.0014	0.0014	0.0014	
	Lifetime	years	30	30	30	30	30	30	30	
	Efficiency	$coeff_{LHV}$	0.701	0.701	0.701	0.701	0.701	0.701	0.701	
Fischer-Tropsch unit	Capex	€/$kW_{FTL,LHV}$	1017	1017	1017	1017	915	915	915	Fasihi et al., (2017)
	Opex fix	€/$kW_{FTL,LHV}$	30.5	30.5	30.5	30.5	27.5	27.5	27.5	
	Opex var	€/$kWh_{FTL,LHV}$	0	0	0	0	0	0	0	
	Lifetime	years	30	30	30	30	30	30	30	
	CO_2 input	$kgCO_2/kWh_{th}$	0.305	0.305	0.305	0.305	0.305	0.305	0.305	
	Efficiency	$coeff_{LHV}$	0.692	0.692	0.692	0.692	0.692	0.692	0.692	
Methanol synthesis unit	Capex	€/$kW_{MeOH,LHV}$	835	835	835	835	835	835	835	Fasihi and Breyer, (2024)
	Opex fix	€/$kW_{MeOH,LHV}$	33.4	33.4	33.4	33.4	33.4	33.4	33.4	
	Opex var	€/$kWh_{MeOH,LHV}$	0	0	0	0	0	0	0	
	Lifetime	years	30	30	30	30	30	30	30	
	CO_2 input	$kgCO_2/kWh_{th}$	0.264	0.264	0.264	0.264	0.264	0.264	0.264	
	Efficiency	$coeff_{LHV}$	0.819	0.819	0.819	0.819	0.819	0.819	0.819	
Ammonia synthesis unit (+ASU & 30 days storage)	Capex	€/$kW_{NH3,LHV}$	1285	1285	1285	1285	1285	1285	1285	Fasihi et al., (2021); Morgan (2013)
	Opex fix	€/$kW_{NH3,LHV}$	64.3	64.3	64.3	64.3	64.3	64.3	64.3	
	Opex var	€/$kWh_{NH3,LHV}$	0	0	0	0	0	0	0	
	Lifetime	years	30	30	30	30	30	30	30	
	Efficiency	$coeff_{LHV}$	0.859	0.859	0.859	0.859	0.859	0.859	0.859	

Table 2
DACCS key specifications.

Technology	Parameter	Unit	2020	2025	2030	2035	2040	2045	2050	Ref.
CO_2 direct air capture unit	Capex	€/(tCO_2·a)	730	481	338	281	237	217	199	Fasihi et al., (2017, 2019)
	Opex fix	€/(tCO_2·a)	29.2	19.2	13.5	11.2	9.5	8.7	8	
	Opex var	€/$kgCO_2$	0	0	0	0	0	0	0	
	Lifetime	years	20	25	25	30	30	30	30	
	Electricity demand	kWh_{el}/tCO_2	250	237	225	213	203	192	182	
	Heat demand	kWh_{th}/tCO_2	1750	1618	1500	1387	1286	1189	1102	
CO_2 transport and storage	Opex var	€/tCO_2	45	45	45	40	36	33	30	Fasihi et al., (2017, 2019)
	Lifetime	years	30	30	30	30	30	30	30	

Table 3
Financial assumptions for primary steelmaking options considered for integrated and split supply chains.

Technology	Parameter	Unit	2020	2025	2030	2035	2040	2045	2050	Ref.
BF+BOF	Capex	€/t_{CS}	442	442	442	442	442	442	442	Lopez et al., (2023a)
	Opex	€/(t_{CS}·a)	103	103	103	103	103	103	103	
	Lifetime	a	40	40	40	40	40	40	40	
	FLH	h/a	8000	8000	8000	8000	8000	8000	8000	
H-DRI-EAF (integrated)	Capex	€/t_{CS}	414	414	414	414	414	414	414	Lopez et al., (2023a)
	Opex	€/(t_{CS}·a)	12.4	12.4	12.4	12.4	12.4	12.4	12.4	
	Lifetime	a	40	40	40	40	40	40	40	
	FLH	h/a	8000	8000	8000	8000	8000	8000	8000	
H-DRI (standalone)	Capex	€/t_{CS}	230	230	230	230	230	230	230	Lopez et al., (2023a)
	Opex	€/(t_{CS}·a)	6.9	6.9	6.9	6.9	6.9	6.9	6.9	
	Lifetime	a	40	40	40	40	40	40	40	
	FLH	h/a	8000	8000	8000	8000	8000	8000	8000	
EAF (standalone)	Capex	€/t_{CS}	184	184	184	184	184	184	184	Lopez et al., (2023a)
	Opex	€/(t_{CS}·a)	5.52	5.52	5.52	5.52	5.52	5.52	5.52	
	Lifetime	a	40	40	40	40	40	40	40	
	FLH	h/a	8000	8000	8000	8000	8000	8000	8000	

optimisations are carried out in two hierarchical resolutions as detailed in Bogdanov et al. (2023). In the first step, Africa was structured into 6 macro regions and 24 regions in the second step, as shown in Fig. 4. In defining the transition pathways, four scenarios were examined under certain policy constraints (Table 4), namely, Best Policy Scenario (BPS), Best Policy Scenario no carbon costs (BPSnoCC), Current Policy Scenario (CPS), and Current Policy Scenario no carbon costs (CPSnoCC). Fischer-Tropsch fuels will incrementally increase their contribution to the transport sector: from 3% in 2030 to 100% by 2050, with steps at 2035 (10%), 2040 (43%), 2045 (77%). Carbon pricing increases stepwise: starting at 28 €/tCO_2 in 2020 and rising to 150 €/tCO_2 by 2050, with increments to 52 €/tCO_2 in 2025, 61 €/tCO_2 in 2030, 68 €/tCO_2 in 2035, and 75 €/tCO_2 in 2040.

RE resource potentials include hourly generation profiles of solar PV, wind power, and hydropower. Region average generation profiles for optimally fixed-tilted PV, single-axis tracking PV, concentrated solar power (CSP), and wind power are calculated based on methods described in (Bogdanov and Breyer, 2016), from generation profiles in 0.45 × 0.45° nodes resolution. Profiles for single-axis tracking PV are calculated according to (Afanasyeva et al., 2018), based on resource

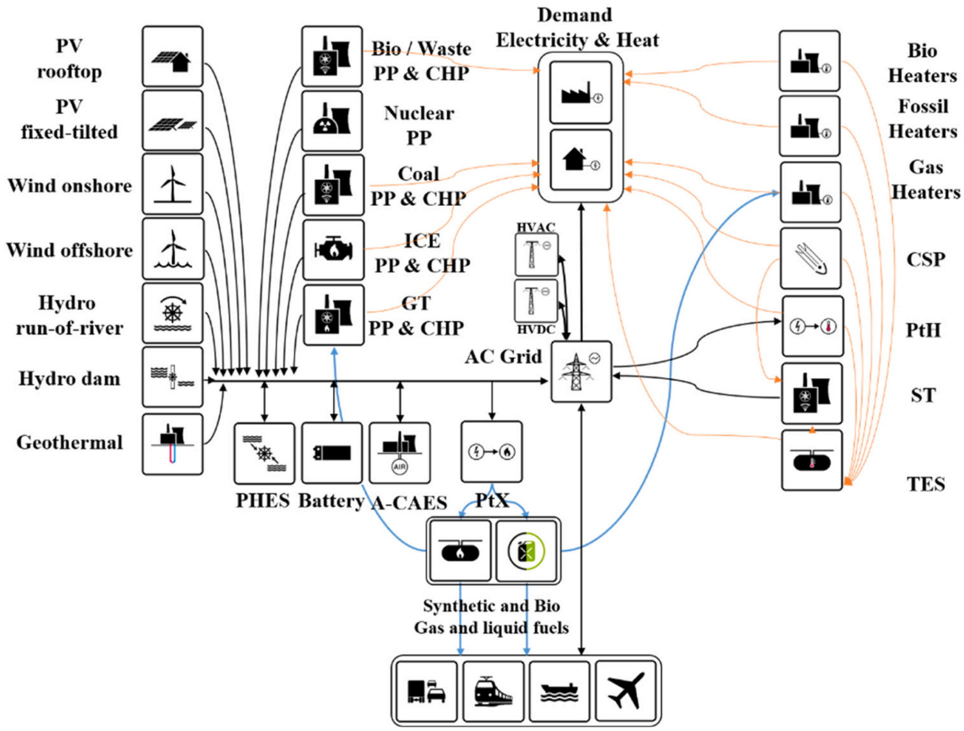

Fig. 1. Schematic of the LUT-ESTM design. Abbreviations: PP, power plant, ST, steam turbines, PtH, power-to-heat, ICE, internal combustion engine, GT, gas turbines, A-CAES, adiabatic compressed air storage, PtG, power-to-gas, PHES, pumped hydro energy storage, TES, thermal energy storage, CHP, combine heat and power, PtX, power-to-X, CSP, concentrating solar thermal power, AC, alternating current, HVAC, high voltage alternating current, HVDC, high voltage direct current. The black arrows denote electricity flow, orange signifies heat, and blue represents PtX.

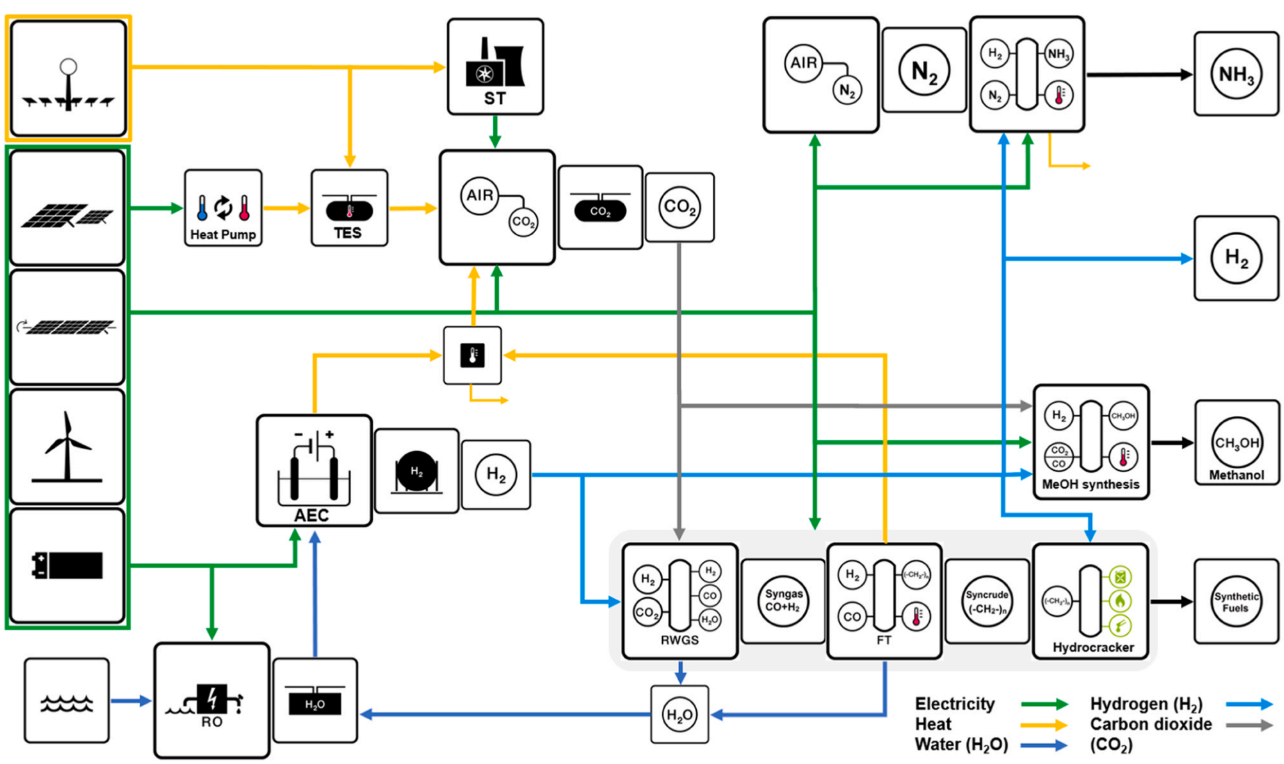

Fig. 2. Schematic of e-hydrogen, e-ammonia, e-methanol, and e-FTL value chain. Abbreviations: Alkaline water electrolyser; AEC, Fischer-Tropsch; FT, reverse osmosis; RO, reverse water–gas shift; RWGS, steam turbine; ST, thermal energy storage; TES. The production and supply chain of e-steel and e-plastic can be found in Lopez et al. (2023a, 2023b)). The diagram is in analogy to ElSayed et al (ElSayed et al., 2023). To enhance readability and facilitate comprehension of the value chain, five distinct coloured arrows have been placed at the bottom right corner of Fig. 2. These arrows represent the flow of electricity, heat, water, hydrogen (H_2), and carbon dioxide (CO_2), guiding the path towards the final PtX products.

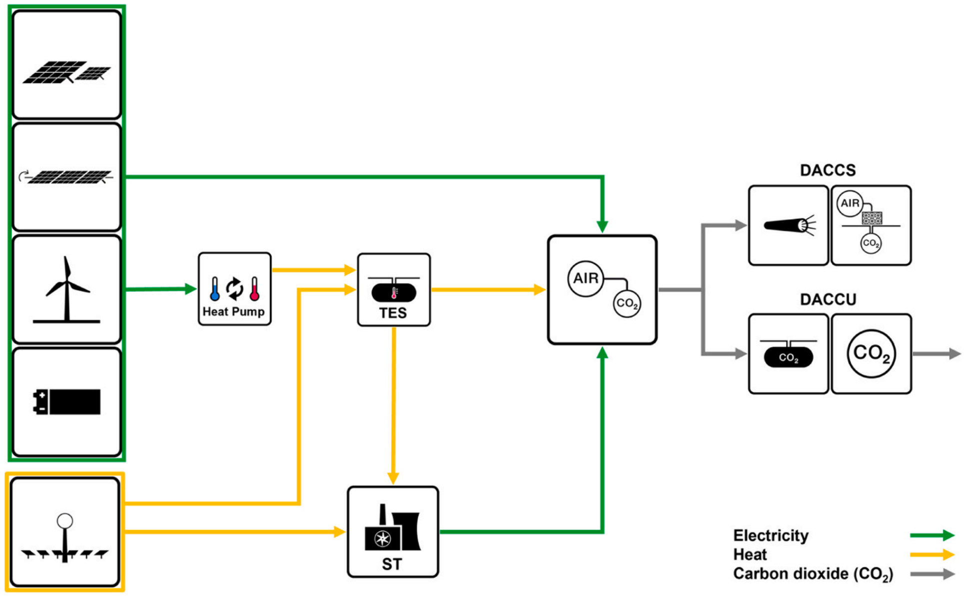

Fig. 3. The PV-battery-HP and CSP-TES CO_2 direct air capture and CO_2 transport and storage value chains for Direct Air Carbon Capture and Utilisation (DACCU) and Direct Air Carbon Capture and Storage (DACCS) applications. Abbreviations: Steam turbine; ST, thermal energy storage; TES. The diagram is in analogy to ElSayed et al. (2023).

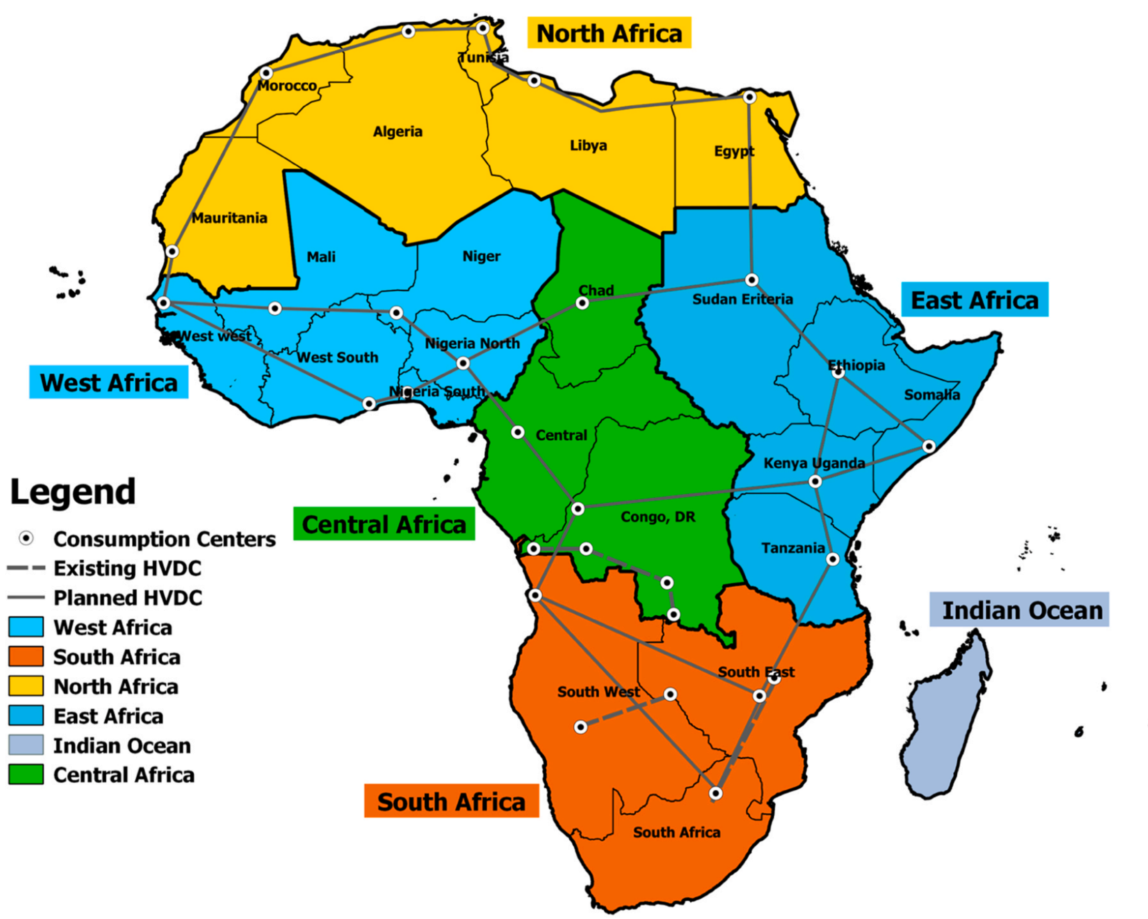

Fig. 4. Overview of Africa structured in 6 macro regions (boldened) and 24 regions. **North Africa** – MA: Morocco; LY: Libya; TN: Tunisia; DZ: Algeria; EG: Egypt; MW: Mauritania, Western Sahara; **West Africa** – WW: Senegal, Sierra Leone, Gambia, Guinea Bissau, Cape Verde Islands, Guinea, Liberia; ML: Mali; WS: Benin, Cote DIvoire, Burkina Faso, Togo, Ghana; NE: Niger; NIG-S: Nigeria South; NIG-N: Nigeria North; **East Africa** – SER: Sudan, Eritrea; ETH: Ethiopia; SOMDJ: Djibouti, Somalia; KENUG: Kenya, Uganda; TZRB: Tanzania, Rwanda, Burundi; **Central Africa** – TCD: Chad; CAR: Sao Tome and Principe, Congo, Guinea Equatorial, Republic of Gabon, Cameroon, Central African Republic; COG: Democratic Republic of Congo; **South Africa** – SW: Angola, Namibia, Botswana; ZAFLS: Lesotho, SE: Malawi, Mozambique, Zambia, Zimbabwe, Swaziland; and **Indian Ocean** – Seychelles, Mauritius, Madagascar, Comoros Islands, and Mayotte.

Table 4
Scenario assumptions for entire Africa.

	BPS	BPSnoCC	CPS	CPSnoCC
Technology				
Fischer-Tropsch liquids (FTL)	Yes		NO	
New fossil fuel and nuclear power plant	No new coal, oil, and nuclear; existing plants are phased out based on lifetime. Gas turbines allowed due to possibilities to switch to sustainable fuels.		New coal, oil, gas, and nuclear plants can be built.	
Hydropower and pumped hydro energy storage (PHES)	Across scenarios hydropower plants and PHES are refurbished every 35 years and never decommissioned			
Policy				
Carbon pricing	Yes	No	Yes	No

data of NASA (Stackhouse and Whitlock, 2008, 2009), reprocessed by the German Aerospace Centre (Stetter, 2012). The hydropower feed-in profiles are computed based on daily resolved water flow data for the year 2005 (Verzano, 2009). The sustainable and economic hydropower potential is obtained from (Gernaat et al., 2017). The potentials for biomass and waste resources were calculated based on the method described in (Mensah et al., 2021) and further classified into categories of solid wastes, solid residues, and biogas. Geothermal energy potential is estimated according to the method described in (Aghahosseini and Breyer, 2020). Fig. 5 shows the feed-in profiles of fixed-tilted PV, single-axis tracking PV, onshore wind, and CSP.

The hourly electricity load profile is calculated as a fraction of the total demand for each sub-region based on synthetic load data weighted by the sub-region's population (Toktarova et al., 2019). The power demand is categorised into residential, commercial, and industrial end-users. Heat demand was divided into domestic hot water heating, biomass for cooking, and industrial processes. The heat demand profiles were generated according to Keiner et al. (2021). Further, the heat demand is classified as low, medium, and high temperatures. The transport sector demand is classified according to Khalili et al. (2019). into road, rail, marine and aviation. The desalination demand is projected as a function of water stress index and total water demand is estimated for each year during the transition, according to Caldera and Breyer (2020).

3. Results

This section presents the results of the various energy pathways for

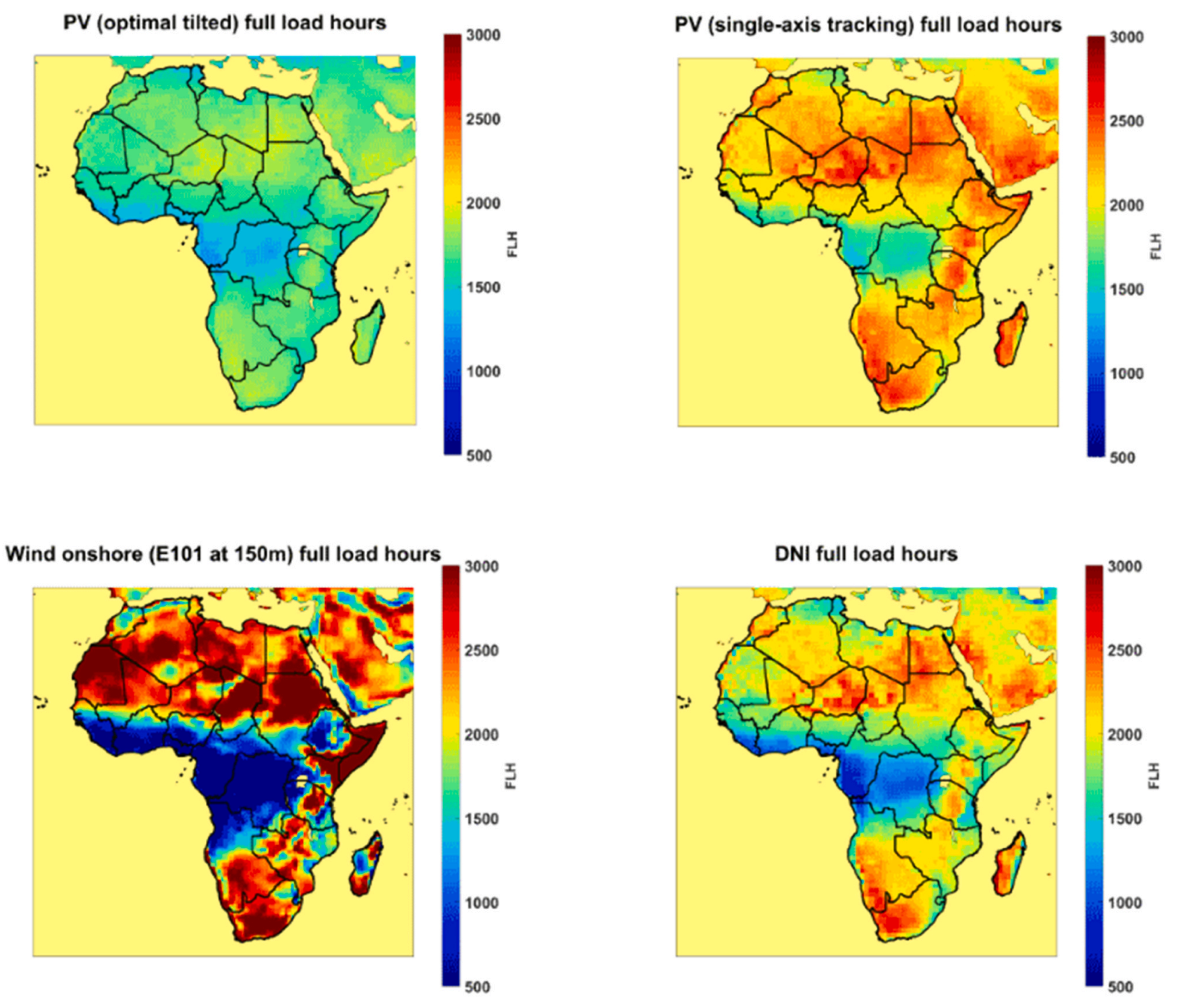

Fig. 5. **:** Maps of Africa showing annual full load hours for optimally tilted (top left) and PV single-axis tracking (top right), wind (bottom left), and CSP solar field (bottom right).

Africa. An overview of primary energy demand, electricity and heat generation outlook, installed capacity for fuel conversion (power-to-X capacities), and costs and export potential of e-fuels, e-chemicals, and e-materials.

3.1. Evolution of primary energy demand through the transition

The primary energy demand increased from around 7555 TWh in 2020 to a range of 8535 – 13,335 TWh in 2050, depending on the scenario, as depicted in Fig. 6. Rapid electrification across the energy-industry system plays a crucial role in driving the energy transition towards a higher share of RE, especially in the BPSs, by a factor of 2–3 compared to the 2020 level. As observed in this study, **low-cost renewable electricity emerges as the main energy carrier for the future**, replacing fossil fuels and unsustainable bioenergy. In the BPS, fossil fuels and unsustainable bioenergy used in the energy sector are substituted by RE technologies directly extracting electricity, in particular solar PV, complemented by wind turbines, hydropower, bioenergy, and geothermal. The trajectory of the primary energy demand through the transition depends on the level of direct renewable-based electrification, which depends on technological adoption, switches in powertrains and the evolution of production processes in the heat, transport, and industry sectors across scenarios. The level of indirect electrification, i.e., the rate of adoption of e-fuels, is another critical factor influencing energy development. Both factors are responsible for the lower levels of primary energy demand in the BPSs compared to the CPSs, as shown in Fig. 6. The rate of electrification increased from around 3% in 2020 to 54 – 92% in the BPSs, and 18 – 22% in the CPSs by 2050. The continuous use of fossil fuels drives the higher level of energy demand in the CPSs.

The primary energy demand by sector is shown in Fig. 7. Across the four scenarios, primary energy demand will increase by 2050 compared to 2020. From a sector perspective, primary energy demand for the power and transport sectors grows across the four scenarios through the transition. The power sector demand grows from about 1550 TWh in 2020–3101 – 3758 TWh by 2050, depending on the scenario. Similarly, the transport energy demand grows from around 1870 TWh in 2020–3373 – 5718 TWh, depending on the scenario. However, primary energy demand for the heat sector declined through the transition in the BPSs but increased in the CPSs. Desalination energy demand is the lowest throughout the transition period and is below 220 TWh across scenarios by 2050.

3.2. Electrification and defossilisation across the energy-industry system

The electricity supply mix across scenarios is dominated by solar PV by 2050, contributing 42–88%, followed by gas turbines with 3–19%, hydropower 2–14%, wind power 5–11%, coal 0–8%, bioenergy 1–2%, and nuclear power 0–2%. Fossil fuel-based electricity generation is highest in the CPS, accounting for 30% of total generation by 2050 and 4% in BPS excluding carbon costs, illustrating the economics of renewables. Total electricity generation remains below 2000 TWh until 2030 and then increases to 3487–10,298 TWh by 2050, depending on the scenario. The BPS has 1.4–2.9 times more electricity generation than the CPS in 2050.

Heat generation increased significantly from nearly 1000 TWh in 2020–1841 – 2224 TWh, depending on the scenario, as shown in Fig. 9. Across the four scenarios, electric heating and bioenergy supply the largest share of heat generation from 2030 onwards. Deep defossilisation of the BPS is achieved via direct and indirect electrification based on renewables. RE fuels supply the hard-to-abate industry heat demand in the BPS. However, CPSs and the BPSnoCC still utilise fossil fuels in the heat until 2050.

Fuel supply in the transport sector differs across scenarios, as depicted in Fig. 10. Fossil fuels remain relevant through the transition in the CPSs, accounting for 94% of transport fuel energy demand by 2050. In the BPSnoCC, fossil fuels dominate in 2050 (32%), followed by grey hydrogen (29%), electricity (28%), and methane (11%). Similar fuel shares are observed in the carbon-neutral BPS scenario, except that grey hydrogen is replaced by green e-hydrogen and fossil fuels are substituted by e-FTL fuels.

The transport, heat and energy-intensive industry sectors can transition to being more sustainable by substituting fossil fuels with e-fuel and e-chemical solutions, as demonstrated in the BPS carbon neutral pathway in Fig. 11. Transitioning to a carbon-neutral energy system is less energy-intensive than the carbon-intensive pathway. The supply of fuels and chemicals decreased substantially from around 7500 TWh in 2020 to a range of 4410 – 4717 TWh in the BPSs by 2050, and contrarily

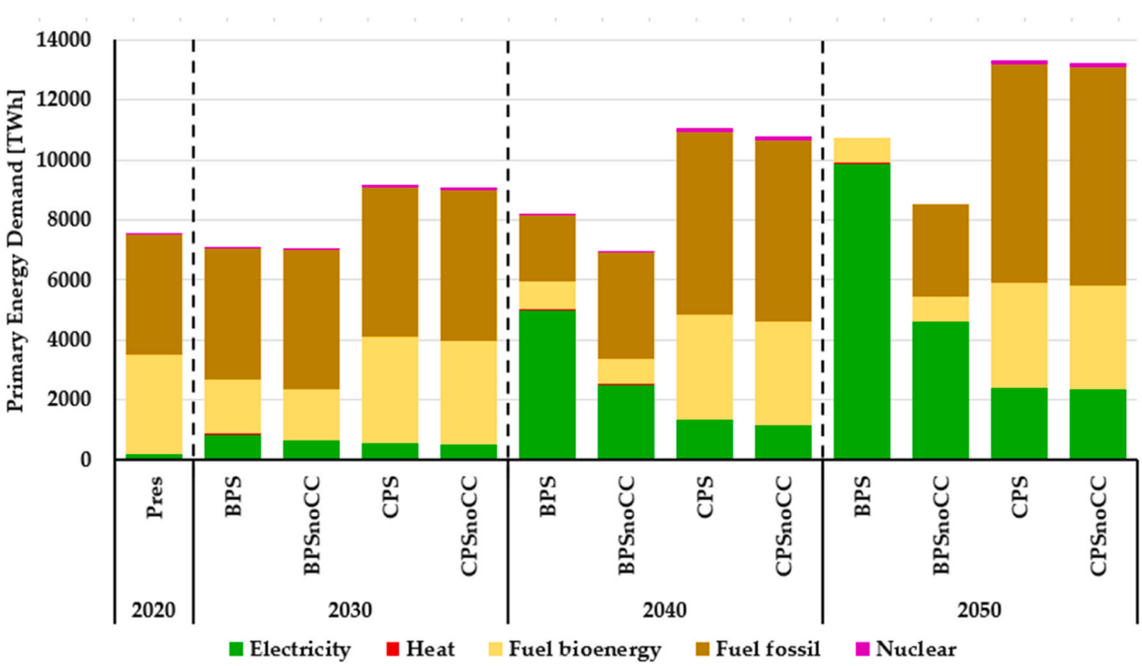

Fig. 6. The development of primary energy demand by fuel-type across all scenarios in 10-year intervals from 2020 to 2050.

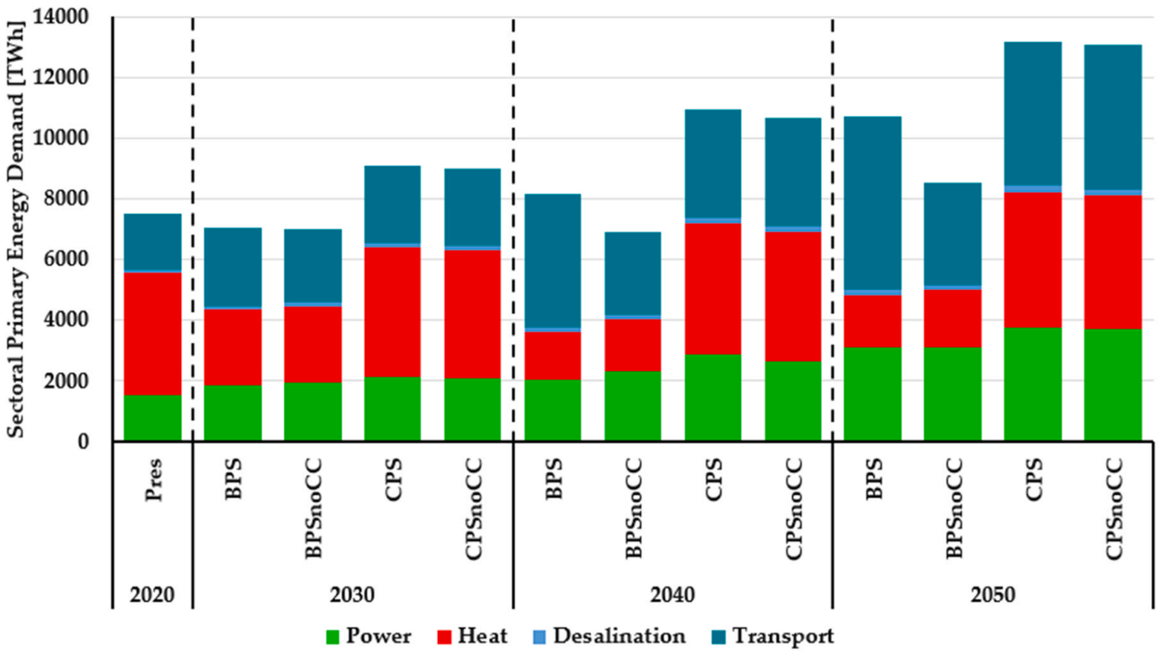

Fig. 7. The development of primary energy demand sector-wise across all scenarios in 10-year intervals from 2020 to 2050.

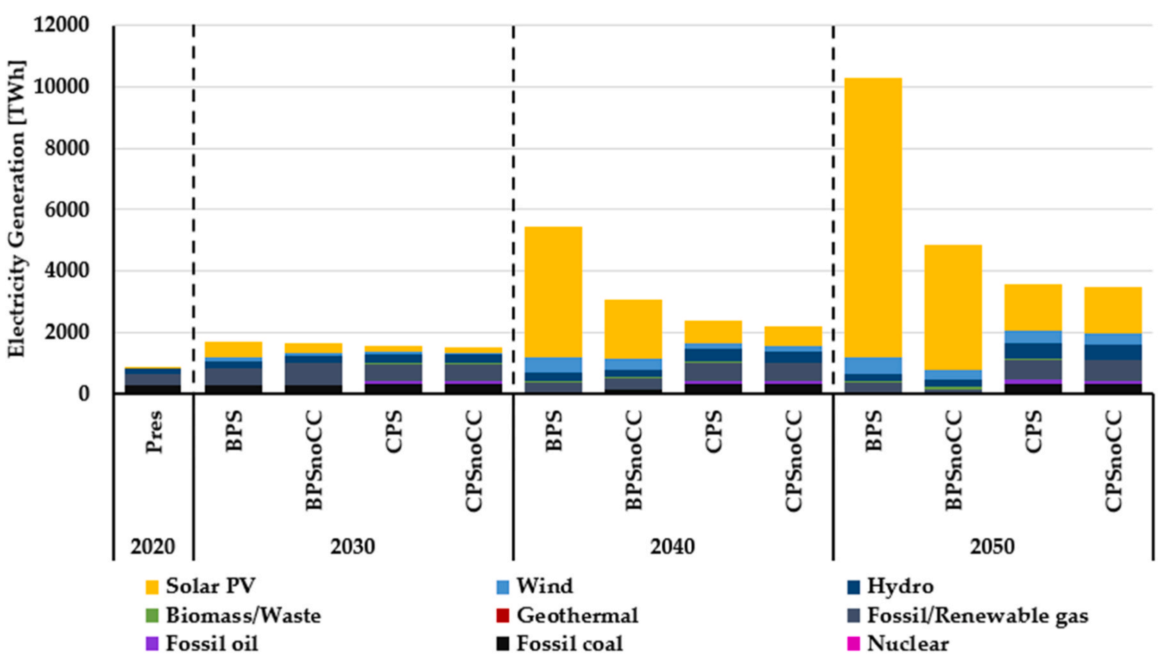

Fig. 8. Electricity generation technology-wise across all scenarios in 10-year intervals from 2020 to 2050.

the value increased to a range of 11,000 – 11,141 TWh in the CPSs. Fuels and chemicals supply is 2.4 times lower in the BPSs compared to the CPSs in 2050. This massive reduction is spurred by substituting fossil fuels and unsustainable bioenergy with RE technologies directly extracting electricity.

Africa's low-cost renewable electricity can be used to produce e-fuels, chemicals, and e-materials. As costs are critical to economies, having access to low-cost renewable electricity could provide vital benefits in the future for new industrial opportunities in producing e-fuels, e-chemicals, and e-materials. The levelised cost of electricity (LCOE) declines through the transition across scenarios, except in the CPSnoCC, as shown in Fig. 12. In 2050, the cost per unit of delivered electricity is 26 €/MWh in the BPS, 34 €/MWh in the BPSnoCC, 76 €/MWh in the CPS, and 47 €/MWh in the CPSnoCC. In the BPSs, low-cost electricity can be attributed to low-cost solar PV. This implies that renewable electricity is set to emerge as the most economical in Africa and will drive a low-cost Power-to-X Economy.

The regional distribution of fuels and chemicals, electrolyser installed capacity, and LCOE is depicted in Fig. 13. The regional distribution of fuels and chemicals and electrolyser installed capacity depends on local demand, full load hours of RE technologies and LCOE. The total supply of fuels and chemicals is 3414 TWh in 2050. Nigeria and Egypt, with 660 TWh and 617 TWh, dominated the overall supply of fuels and chemicals, representing 19% and 18% of the regional supply. The regional hydrogen utilisation is 4404 TWh in 2050. Egypt, with 1079 TWh, has the highest regional hydrogen utilisation representing

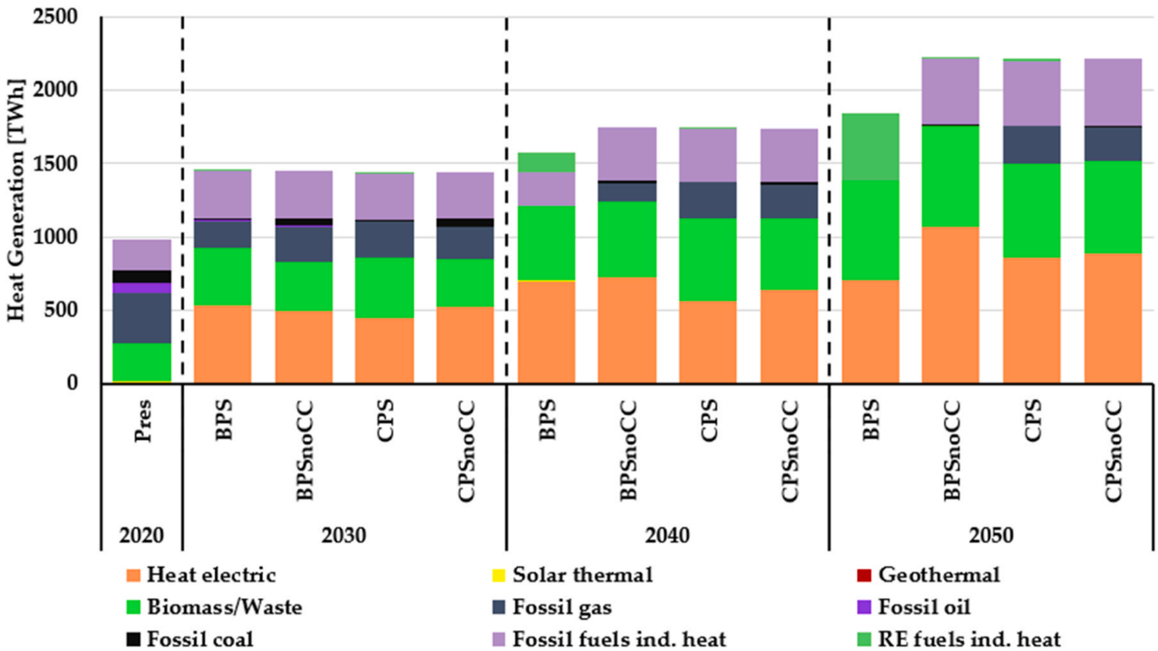

Fig. 9. Technology-wise heat generation trajectory across all scenarios in 10-years interval from 2020 to 2050.

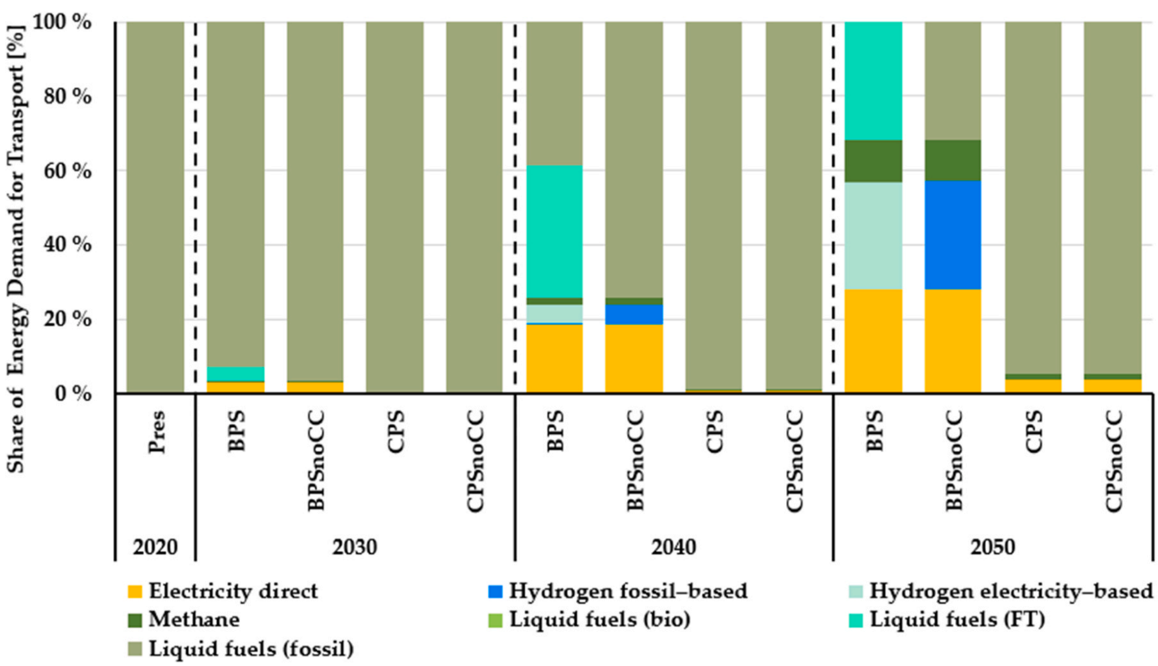

Fig. 10. Structure of the final energy demand in the transport sector in 10-year intervals from 2020 to 2050.

25% of total African production, followed by Nigeria with 871 TWh (20%) and South Africa with 211 TWh (4.8%). Electrolysers play a critical role in the power-to-X processes, as hydrogen is an essential precursor in producing e-fuels, e-chemicals, and e-materials. The total installed electrolysers capacity is projected to 1469 GW in 2050. The highest installed capacity of electrolysers can be found in Egypt, Nigeria, and South Africa. The regional average LCOE is 26 €/MWh, comprised of generation, storage, curtailment, and grid costs. Mali and Nigeria-North have the lowest LCOE in 2050 at around 14 €/MWh and 16 €/MWh. The highest average LCOE is found in Tunisia and Chad at around 61 €/MWh and 50 €/MWh, respectively, by 2050. In 12 sub-regions, LCOE is below the regional average. Many coastal African countries with low-cost renewable electricity are potential sites for low-cost green e-fuels, e-chemicals, and e-materials export. The export potential of green e-fuels, e-chemicals, and e-materials from selected countries and the entire continent is explored in subsequent sections.

3.3. e-Fuel, e-chemical, e-steel, and e-plastic cost trajectory and export potential

The cases presented in this study on PtX in Africa emphasise the importance of diverse perspectives. Examining PtX products for both domestic and export markets, Egypt focuses on e-fuels and e-chemicals, while Morocco analyzes e-steel and e-plastic. Trading in feedstock and

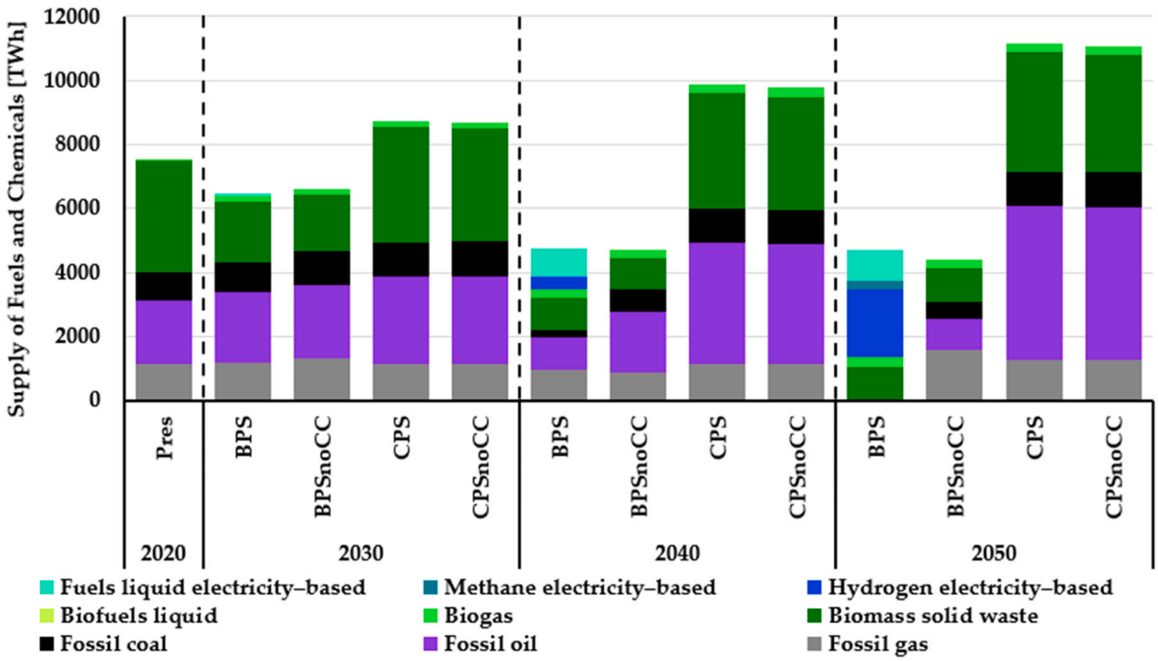

Fig. 11. Supply of fuels and chemicals through the transition across scenarios in 10-year intervals from 2020 to 2050.

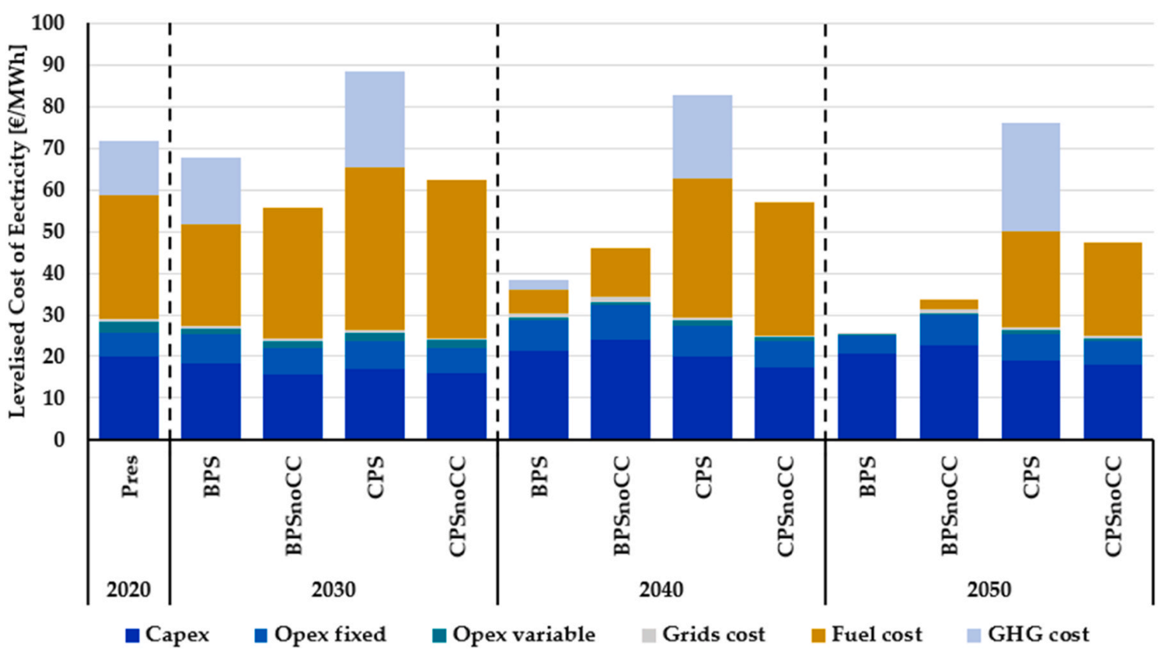

Fig. 12. Levelised cost of electricity through the transition across scenarios in 10-year intervals from 2020 to 2050.

final products is prioritised over hydrogen due to cost considerations. Initial benchmarking considers local production costs and imports from low-cost sites. The results underscore Africa's potential for a thriving export industry in PtX products, supported by affordable renewable electricity, primarily solar PV. Each case study identifies unique market dynamics.

3.3.1. African export competitiveness of all e-fuels in a global perspective

Growing global demand for e-fuels and e-chemicals in the future may lead to increased global trade and attract new participants to the energy markets. Africa, with its abundant solar and wind resources, has a potential to emerge as one of the major exporters of green e-fuels and e-chemicals. In 2050, driven by low e-fuel and e-chemical production costs, Africa may become one of the export-oriented regions that will globally trade e-LNG, e-FTL, e-ammonia, and e-methanol (Galimova et al., 2023a).

In terms of costs, the levelised cost of e-methane is some of the lowest in Africa, especially in sub-Saharan Africa and can be as low as 114–130 €/MWh in 2030, compared to the average levelised cost of fuel (LCOF) in Europe of 133 €/MWh. By 2050, the costs are projected to decline to 44–55 €/MWh, whereas LCOF in Europe is 64 €/MWh. LCOF of e-FTL is higher ranging from 160–230 €/MWh in 2030 and is projected to decline to as low as 71 €/MWh by 2050 compared to an average LCOF of 96 €/MWh in Europe. e-Ammonia and e-methanol costs are in the range of 105–135 €/MWh and 118–153 €/MWh in 2030, respectively, and 44–52 €/MWh and 48–55 €/MWh in 2050 compared to 53–72 €/MWh and

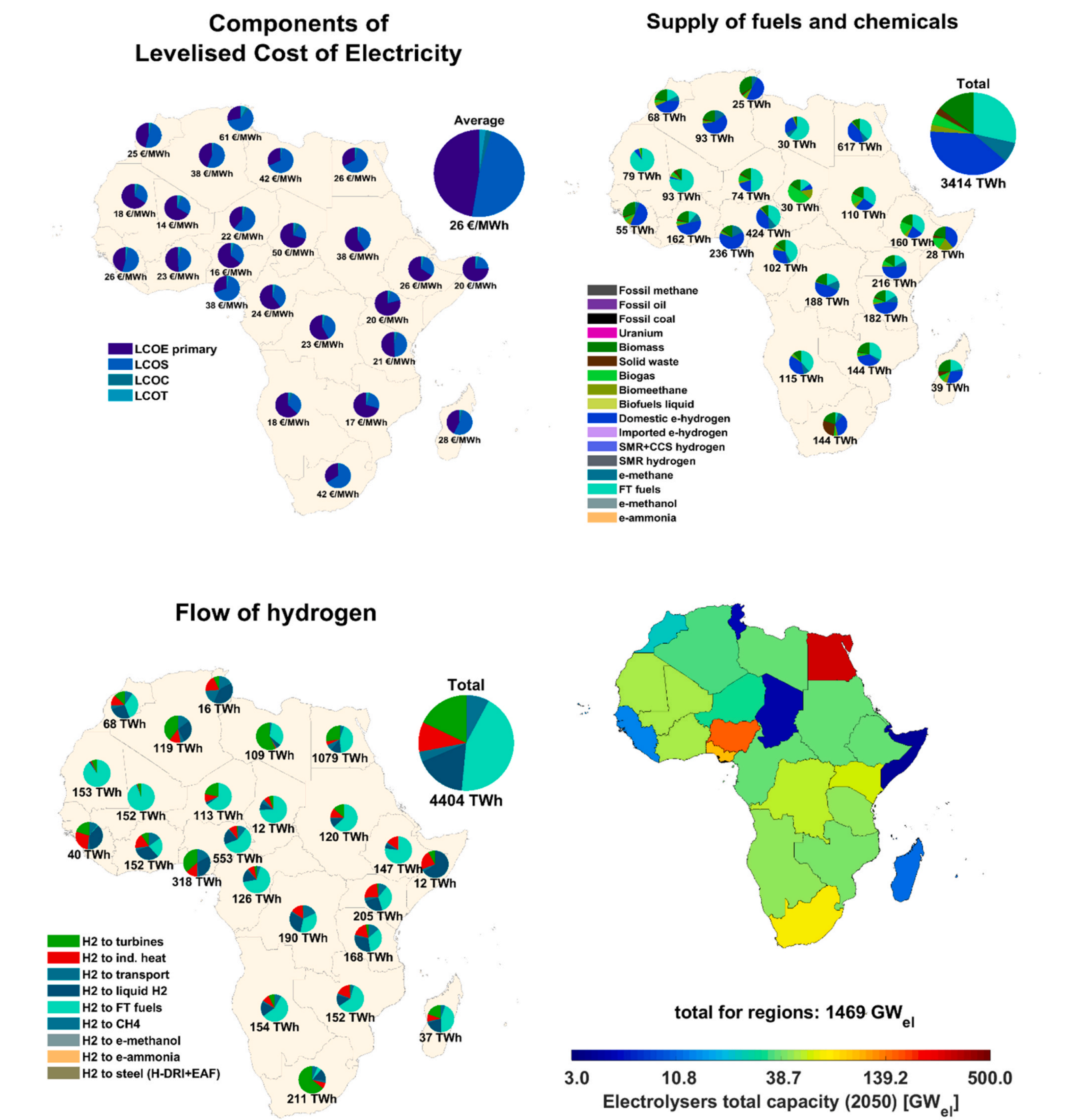

Fig. 13. Regional distribution of LCOE (top left), supply of fuels and chemicals (top right), hydrogen utilisation (bottom left) and electrolysers capacity (bottom right) in 2050 in the BPS.

59–80 €/MWh in Europe in 2050, respectively.

As shown in Fig. 14, the exported volumes in 2030 are relatively low, standing at around 38 TWh, but the volume grows more than 100-fold to nearly 4500 TWh of e-fuels and e-chemicals in 2050. Africa's role as an exporter changes considerably from a negligible supplier of energy carriers in 2030, representing less than 3% of total global exports, to a major exporter accounting for 47% of total traded volume (Galimova et al., 2023a). e-Methane and e-FTL mostly originate from Angola, Namibia (Creamer Media's. Namibia signs agreement opening way for feasibility study of $10bn green ammonia project, 2023), Tanzania, and Nigeria. The same regions export e-ammonia, additionally supported by Zimbabwe and Mozambique, whereas e-methanol is mostly exported from Nigeria and several of its neighbouring countries.

However, taking into account the continent's vast RE resources, Africa's potential for exports is even higher. In 2050, 4600 TWh of e-methane, 7000 TWh of e-FTL, 3300 TWh of e-ammonia, and 11,700 TWh of e-methanol can be produced and exported from Africa. These volumes exceed the total global demand for these e-fuels and e-chemicals.

Production of large volumes of e-hydrocarbons (e-methane, e-FTL, e-methanol, e-carbon fibres) will also depend on the availability of CO_2 as a raw material. The total demand for CO_2 has been estimated to reach

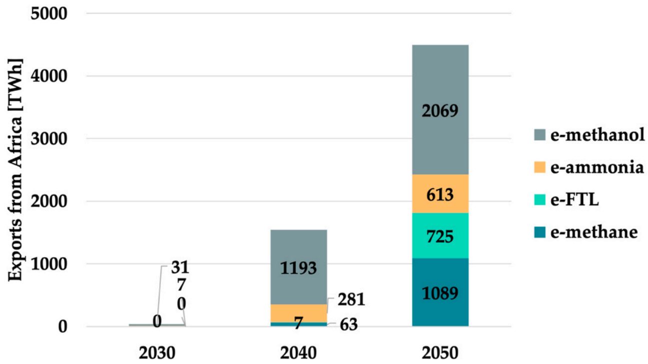

Fig. 14. Total exported volumes of e-fuels and e-chemicals from Africa to other regions of the world in 2030, 2040, and 2050.

634 $MtCO_2$ by 2030, with most of it concentrated in East Asia, Western Europe, and North America, whereas demand in Africa is quite limited (Galimova et al., 2022). By 2050, the demand is estimated to grow 10-fold, reaching 6076 $MtCO_2$, most of which is again concentrated in the same regions, but with increased demand also across African countries, where the total demand for CO_2 is projected to be 552 $MtCO_2$. Some of the demand can be covered by sustainable and unavoidable point sources such as pulp & paper mills, cement plants, and waste incinerators, that cumulatively can supply 78% of the total demand for CO_2 as a raw material in Africa, while the rest of CO_2 can be sourced from DAC. Most of the CO_2 in Africa is projected to be supplied from cement plants in 2030 and 2040, as cement production is projected to grow significantly in the future as the region undergoes industrialisation and rising standards of living with demand for expanded infrastructure and buildings. Supply of CO_2 from waste incinerators is projected to be negligible in 2030 but is estimated to become the dominant source by 2050 across much of Africa, as the population and waste volumes per capita grow (Galimova et al., 2022).

In addition, it is important to consider that economic viability of exporting these e-fuels and e-chemicals to other countries depends not only on the fuel costs in importing countries, but also on the type of fuel and associated transportation costs. For example, hydrogen trading may not be attractive due to additional high transportation costs (Galimova et al., 2023c), driven mainly by expensive pipelines and high liquefaction costs. Morocco, due to its lower e-hydrogen costs compared to Germany or Finland and its proximity to Europe, could be an attractive supplier of e-hydrogen to Europe. Nevertheless, after accounting for transportation costs, the final cost of imported e-hydrogen may be 35–45% higher compared to the cost of local production as shown in Fig. 15.

When considering the trading of other e-fuels, lower transportation costs may promote exports from Morocco and other African countries. It has been shown that e-ammonia exports from Morocco to Germany and Finland can be economically viable if the chemical is shipped by sea, but not via pipelines (Galimova et al., 2023b).

3.3.2. *Case Egypt: e-fuel and e-chemical export potential*

To investigate the Egyptian e-fuels and e-chemicals domestic and export potential to Europe, five scenarios were examined (see Table 5) and the levelised cost of e-fuels and e-chemicals were calculated across the supply chain. These costs were estimated for e-hydrogen, e-ammonia, e-methanol, and e-FTL, as shown in Fig. 16, as well as their respective domestic and export volumes. Across the modelled scenarios,

Table 5
Scenario description for the case of Egypt.

Scenario name	Description
Best policy scenario 4% high transmission (BPS−4-HT)	This scenario aims for 100 % RE and zero CO_2 emissions by 2050 for the entire energy system: electricity, heat, transport, and desalination. Throughout the transition period, there is an annual growth limit for the share of installed capacity RE of 4%. Inter-zonal electricity transmission is unrestricted. Production of e-fuels, including e-FTL, is allowed for domestic consumption. CO_2 emission costs are levied from 2020 to 2050.
Best policy scenario with e-fuels/e-chemicals exports (**BPS-E**)	This scenario corresponds to **BPS −4-HT**, but with additional exports of e-fuels and e-chemicals. Egypt is assumed to supply 10% of Europe's demand for e-hydrogen, e-ammonia, e-methanol, and e-FTL in the period 2025–2050.
Best policy scenario with e-fuels/e-chemicals exports and CDR services via CSP (**BPS-EC-CSP**)	This scenario combines the features of BPS−4-HT and BPS-E, with CO2 removal (CDR) services (via DACCS) fulfilling 5% of global CDR requirements from 2035 onwards, using CSP with TES.
Best policy scenario with e-fuels/e-chemicals exports and CDR services via solar PV (**BPS-EC-PV**)	This scenario is the same as **BPS-EC-CSP** but using solar PV with batteries and heat pumps.

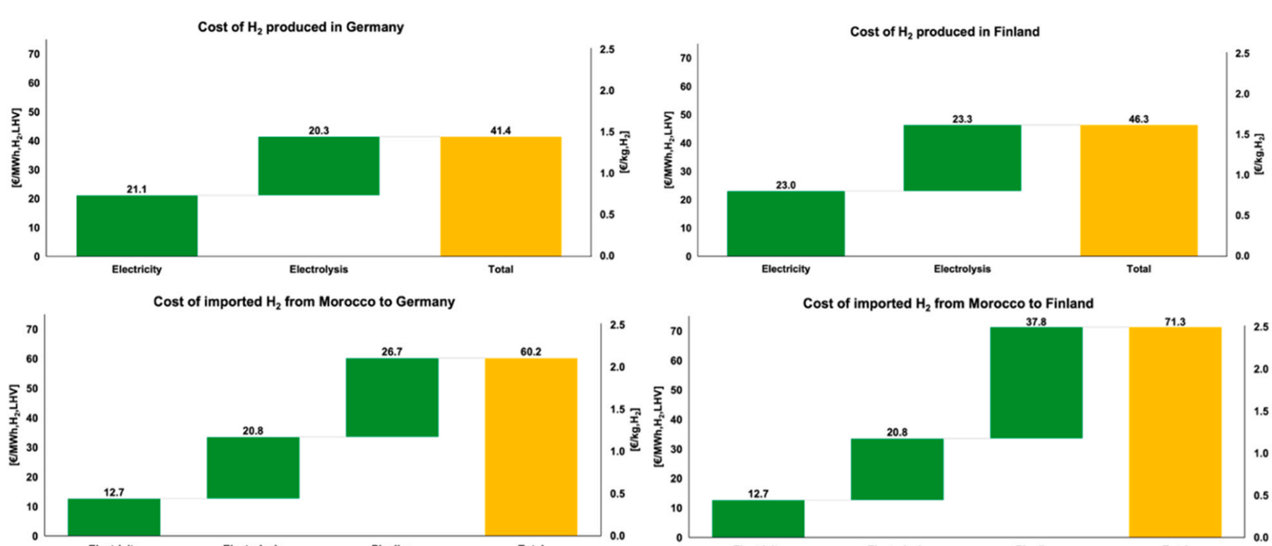

Fig. 15. Cost of e-hydrogen produced locally (top) in Germany (left) and Finland (right) and imported from Morocco (bottom) in 2050.

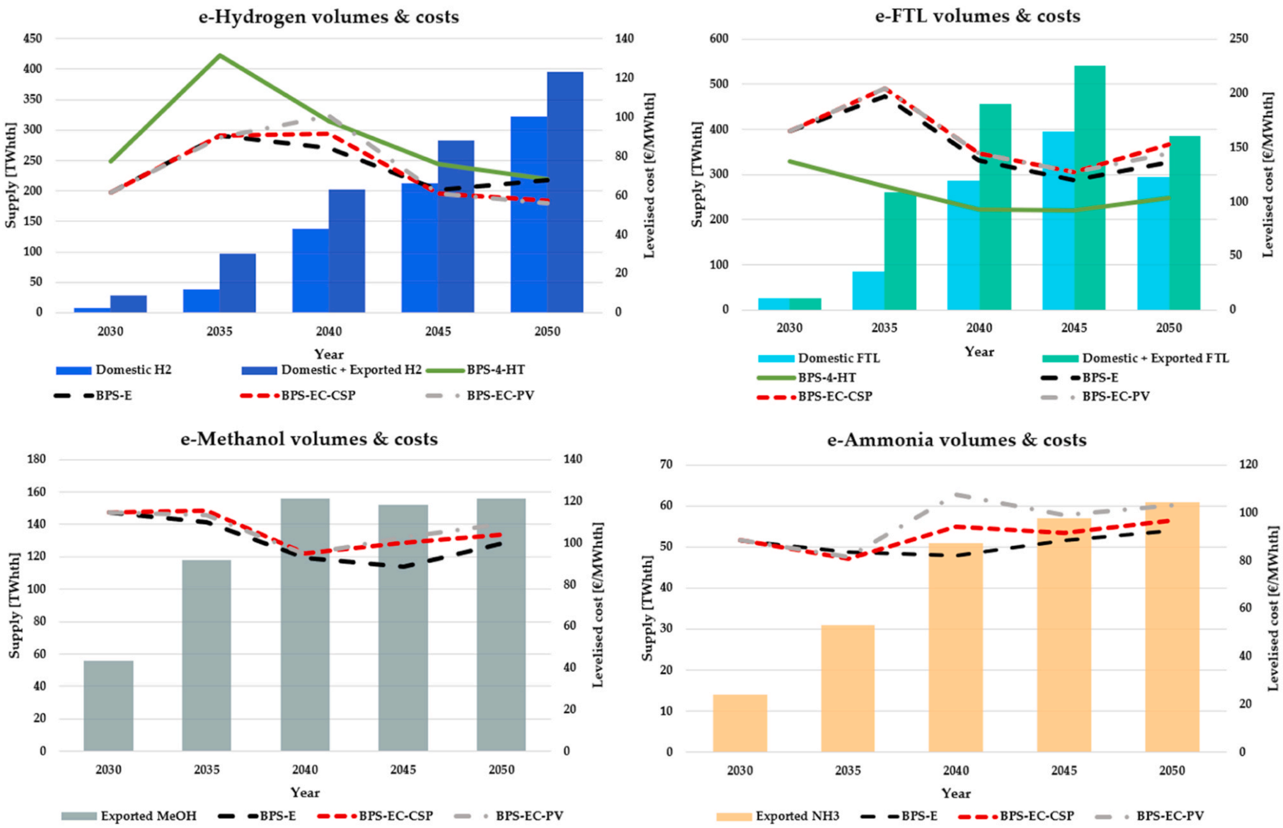

Fig. 16. Cost of e-hydrogen produced locally (top left), cost of e-FTL (top right), cost of e-methanol (bottom left) and cost of e-ammonia (bottom right) through the transition.

various conditions (see Table 5) are examined to assess the effects of supplying CDR services and exporting e-fuels/e-chemicals on energy system performance and costs. The **BPS-4-HT** serves as the baseline scenario for comparing energy system performance and costs. The BPS-E focuses on e-fuels/e-chemicals, while the other two scenarios explore the impacts of exporting these products. Additionally, **BPS-EC-CPS** incorporates CPS and TES, whereas **BPS-EC-PV** integrates solar PV with battery storage and heat pumps. The objective is to evaluate and contrast the technical and economic performance of these two system configurations.

In 2035, when large-scale production begins, hydrogen's levelised cost (LCO) peaks but declines as renewables and economies of scale gain traction. Compared with the BPS-4-HT scenario, the LCO hydrogen peak is about 30% lower in scenarios with e-fuels and e-chemicals exports. At 55.9 €/$MWh_{th,LHV}$, the LCO hydrogen for the BPS-EC-PV scenario in 2050 is 18% lower than for the BPS-4-HT.

As the production scale increases, the LCO e-FTL also decreases. However, scenarios with e-FTL export have a higher LCO e-FTL than the BPS-4-HT, which has a LCO e-FTL of 103 €/$MWh_{th,LHV}$ in 2050. In scenarios with e-FTL exports, the LCO e-FTL is 136 €/$MWh_{th,LHV}$ in 2050. Similarly, increasing supply also reduces LCO e-methanol. The least cost of e-methanol is found in the BPS-E, with a significant decline from 153.8 €/$MWh_{th,LHV}$ in 2025 to 99.7 €/$MWh_{th,LHV}$ by 2050. Compared with other e-fuel and e-chemical costs, LCO e-ammonia did not decrease as the production scales. However, at 92.4 €/$MWh_{th,LHV}$ in 2050, the BPS-E delivers the lowest LCO e-ammonia.

3.3.3. *Case Morocco: e-steel and e-plastic exports from Morocco*

To investigate the African e-steel and e-plastic export potential to Europe, the levelised cost of e-steel (LCOCS) and e-polyethylene (LCOPE) were calculated across the supply chain. These costs were specifically calculated for e-steel and e-polyethylene (e-PE) used in Germany, and potentially imported from Morocco.

For the steel production route, steel can be produced either via the conventional blast furnace and basic oxygen (BF+BOF) furnace route, or by using e-hydrogen via the hydrogen direct reduction and electric arc furnace (H-DRI-EAF) route, which are described in detail in (Lopez et al., 2022). In the H-DRI-EAF route, imports can occur as e-hydrogen feedstocks, hot briquetted iron (HBI), or as the final crude steel (CS) product (Lopez et al., 2023a; Devlin and Yang, 2022; Trollip et al., 2022). Fig. 17 shows the LCOCS of local steel production in Germany compared to e-hydrogen, HBI, and e-steel imports from Morocco.

While e-steel for all supply chains is higher in cost than the conventional BF+BOF route in 2020, this trend changes as early as 2030, as HBI and CS imports are 69% and 64% of BF+BOF production in Germany. Compared to local H-DRI-EAF LCOCS in Germany, imported HBI and CS from Morocco are 99.7% and 93%, suggesting that these supply chain routes become the least cost e-steel supplies as early as 2030. As the transition continues, the economic outlook for e-HBI and e-CS exports from Morocco compared to local e-steel in Germany becomes increasingly positive; however, the same cannot be said for e-hydrogen exports. Although e-hydrogen production from hybrid PV-wind power plants can reach levelised costs of hydrogen (LCOH) of 16.5, 13.0, and 10.2 $MWh_{H2,LHV}$ in 2030, 2040, and 2050, respectively, e-hydrogen transportation via large scale pipelines increases the cost of delivered hydrogen by 52.8, 42.3, and 38.0 $MWh_{H2,LHV}$ in 2030, 2040, and 2050, respectively. Comparatively, the transport cost of HBI and CS are quite low, ranging from 9.9–12.6 €/tCS throughout the transition. HBI imports therefore appear to be a more effective use of the low e-hydrogen production costs in Morocco, and simultaneously allow for continued steel production in Germany.

The excellent solar resources in Morocco lead to lower electricity and e-hydrogen production costs compared to Germany, and, thus, the supply chain with lowest LCOCS is CS imports from Morocco. Compared

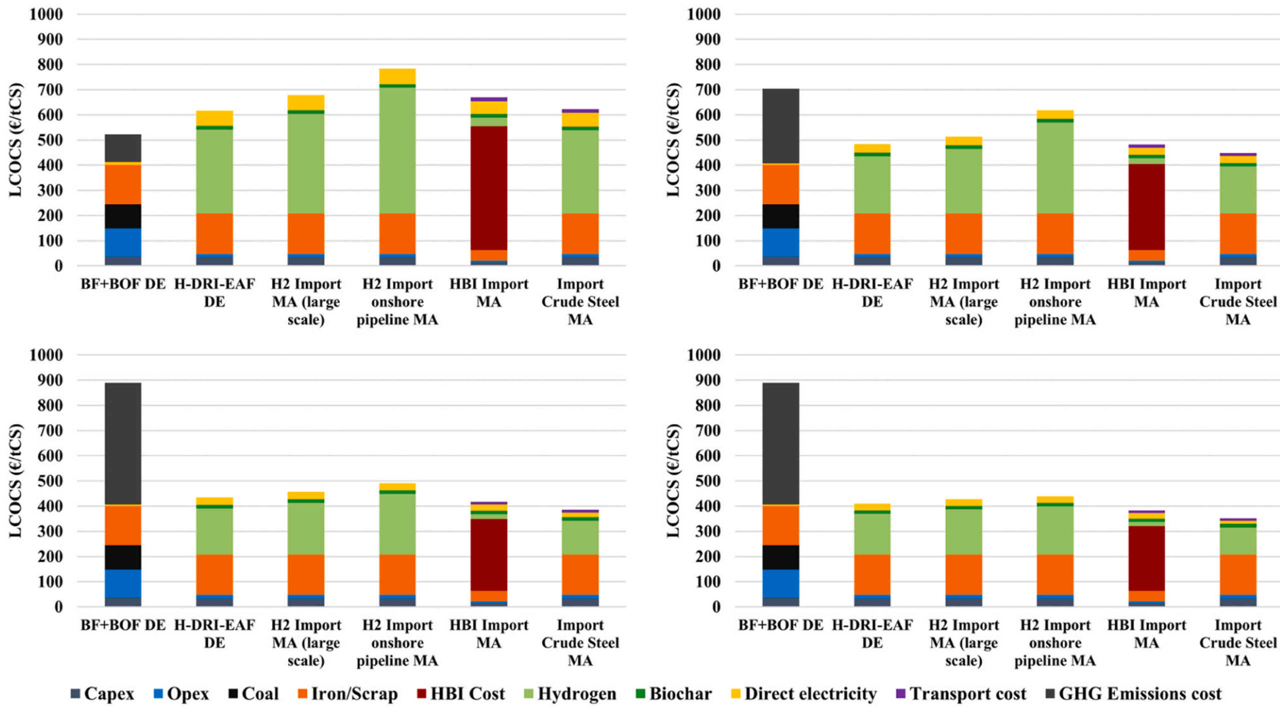

Fig. 17. Levelised cost of crude steel (LCOCS) by supply chain configuration from Morocco (MA) to Germany (DE) in 2020 (top left), 2030 (top right), 2040 (bottom left), and 2050 (bottom right).

to the LCOCS in Germany of 484, 434, and 410 €/tCS in 2030, 2040, and 2050, the LCOCS of CS imported from Morocco is 449, 385, and 351 €/tCS in 2030, 2040, and 2050, respectively. Importantly, the LCOCS of CS imports from Morocco become cheaper than BF+BOF production in Germany in 2030 with GHG emissions costs, and cheaper than BF+BOF production without GHG emissions costs in 2040. Therefore, African exports along the steelmaking supply chain to Germany, and Europe as a whole, particularly HBI or CS exports, may be attractive options to defossilise both African and European steelmaking.

A similar power-to-X supply chain that may experience splits in the future is the e-chemical/e-plastic supply chain. Along the e-plastics supply chain, described in (Lopez et al., 2024; Leppäkoski et al., 2023), e-hydrogen, e-methanol, and e-PE can be exported using the methanol-to-olefins (MTO) route, which can be expected as the primary

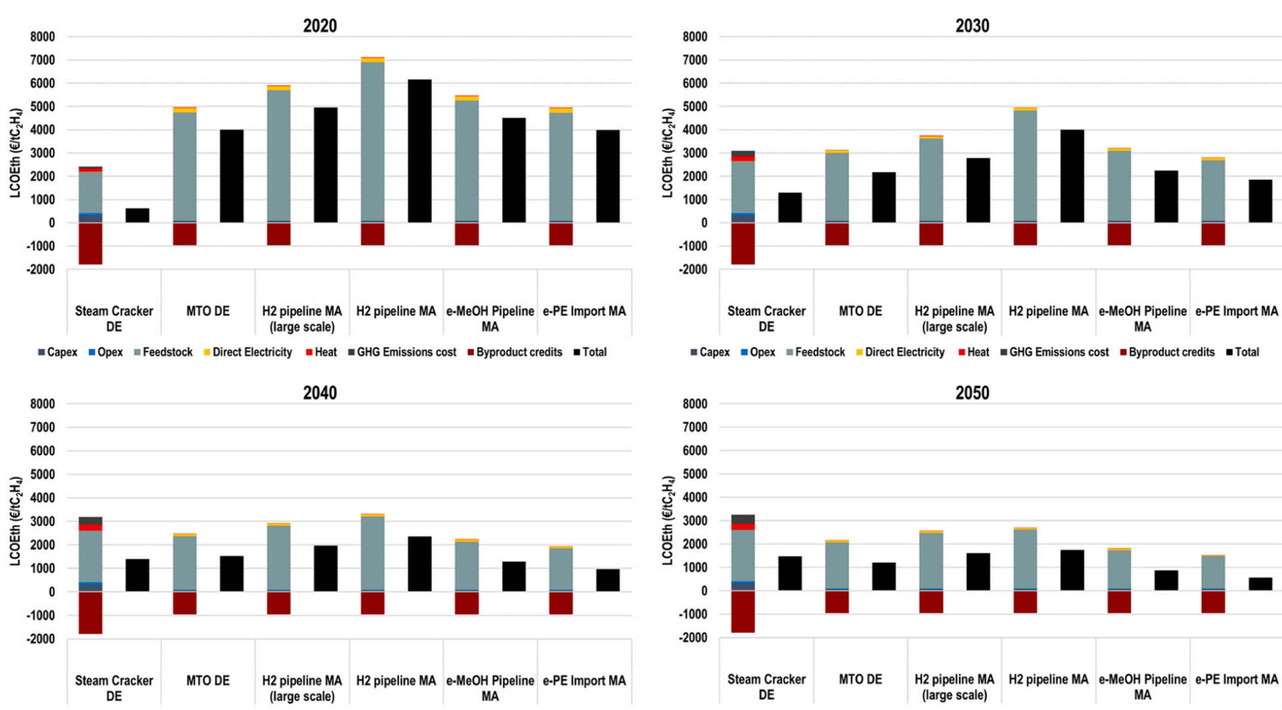

Fig. 18. Levelised cost of ethylene (LCOEth) by supply chain configuration from Morocco (MA) to Germany (DE) in 2020 (top left), 2030 (top right), 2040 (bottom left), and 2050 (bottom right).

defossilised ethylene production route (Lopez et al., 2023b). For the split e-plastic supply chain, CO_2 direct air capture was assumed to supply the CO_2 for methanol synthesis; however, sustainable point source captured CO_2 can be used from e.g., pulp and paper mills and municipal waste incinerators (Galimova et al., 2022). Similar to the e-steel supply chain configurations, the e-plastic supply chains were compared to a reference steam cracker in Germany using naphtha as a feedstock, which is the most widely used European ethylene feedstock (Cefic, 2020).

The results for e-ethylene production, shown in Fig. 18, show that e-ethylene production, and thus, e-PE will be competitive with fossil PE production if African exports are utilised for e-methanol or e-PE. Compared to fossil steel production, the steam cracker has relatively low emissions, as most carbon input is converted to a wide array of hydrocarbon products; therefore, most emissions result from process heating, which was assumed to be supplied by natural gas. Steam cracking of naphtha also has a wider array of co-products, thus resulting in higher byproduct credits compared to the MTO route. Considering these factors, the availability of low-cost renewable electricity to produce e-methanol feedstocks becomes increasingly apparent. With increasing GHG emissions costs and naphtha feedstock prices, the levelised cost of ethylene (LCOEth) increases from 627 €/tC_2H_4 in 2020 to 1475 €/tC_2H_4 in 2050. Conversely, the e-ethylene production decreases from a rather high starting point at 2198 €/tC_2H_4 in 2030 to 1229 €/tC_2H_4 in 2050. During the same period, however, the LCOEth of e-MeOH imports and e-PE imports are 104% and 85%, 85% and 67%, and 75% and 48% of e-ethylene production costs in Germany in 2030, 2040, and 2050, respectively. e-Hydrogen imports consistently lead to the highest LCOEth, as the H_2 pipelines add 28.1–32.3 €/$MWh_{H2,LHV}$ to the LCOH in 2050, depending on the capacity of the pipeline.

The LCOPE, shown in Fig. 19, largely follows the LCOEth trends. As early as 2030, the e-PE imports become the least cost e-PE supply chain configuration considered at 1975 €/tLDPE and become the least cost PE choice in 2040 and 2050 at 1074 and 655 €/tLDPE, respectively. Fig. 19 further shows the distribution of investments and co-product production inside and outside of the EU. Importantly, naphtha steam crackers primarily use naphtha imported from outside of the EU; therefore, feedstock imports can be considered as the default market condition for European chemical production. From this perspective, Morocco may be well positioned to adopt a role of sustainable feedstock supplier to the EU. To successfully move into this role, significant investments will be required in Morocco to develop e-methanol production capacities, which can also serve to defossilise local chemical production. In today's market, PE is also a highly traded commodity (Barrowclough et al., 2020); thus, for PE exported to Europe, a significant opportunity exists for the countries of Africa to supply low-cost e-PE to Europe and benefit from the revenues gained from the co-products of the MTO process.

4. Discussion

This research demonstrates the role developing regions like Africa with abundant RE resources can play in a global decarbonisation strategy in line with the Paris Agreement on climate change. This section is structured into three subsections. The first subsection discusses the **intrinsic value of low-cost renewable electricity in achieving carbon neutrality**. Section 4.2 discusses the **relevance of the Power-to-X Economy for Africa**. Finally, Section 4.3 discusses **the role of Africa in the global defossilisation strategy**.

4.1. Renewable electricity and e-hydrogen – the goldmine of the carbon-neutral energy system

The research findings show that **renewable electricity**, supplied mainly by **solar PV, will emerge as the prime source of low-cost electricity for Africa's future energy systems** due to excellent resource conditions across the continent and the attractive cost decline of the technology (Oyewo et al., 2022; Vartiainen et al., 2020; Jaxa-Rozen and Trutnevyte, 2021; Creutzig et al., 2017; Haegel et al., 1979). Globally, solar PV is expected to play a more significant role, especially considering that countries in the Global South are mainly located in the sunbelt (Oyewo et al., 2022; Jaxa-Rozen and Trutnevyte, 2021; Creutzig et al., 2017; Jacobson et al., 2019a). In several developing countries, the share of solar PV in electricity generation is already

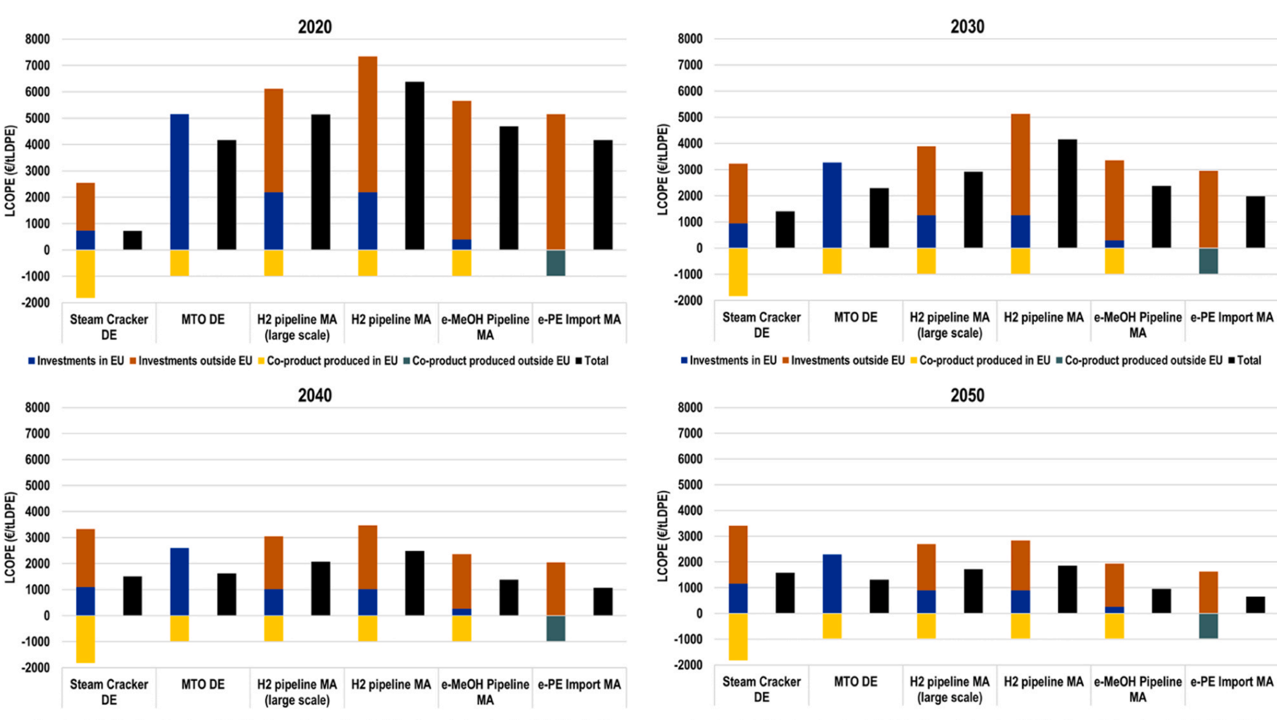

Fig. 19. Levelised cost of polyethylene (LCOPE) by supply chain configuration from Morocco (MA) to Germany (DE) in 2020 (top left), 2030 (top right), 2040 (bottom left), and 2050 (bottom right).

significant, with off-grid solar being an essential technology for rural areas (Jäger-Waldau, 2019; Sovacool et al., 2019). In addition, countries in the Global South, including Africa, Southeast Asia, and South Asia, will see a significant increase in electricity demand as their economies grow and the need for defossilisation increases (Keiner et al., 2023b; Breyer et al., 2023b). As a result, many developing and emerging regions, such as Africa, are greenfields for developing energy systems with high shares of renewables from the outset (Oyewo et al., 2023; Sterl, 2021). This provides an opportunity to leapfrog over costly fossil fuels into sustainable development (Ram et al., 2022a), which countries in the Global North did not have during their past phases of economic development. **Batteries emerge as the key storage solution**, pushing solar PV to the foremost of Africa's future energy system, thus validating the PV-battery hybrid economics and enhancing the Power-to-X Economy, shown in Fig. 20. Recent studies have highlighted the possibility of PV-battery hybrid systems dominating the future energy systems based on low-cost batteries (Vartiainen et al., 2020; Victoria et al., 2021; Schmidt et al., 2017; Kittner et al., 2017; Ziegler and Trancik, 2021). The PV-battery hybrid systems will enable an accelerated transition, rapid electrification, and decentralised variable RE ramping. Innovation and continuous cost reduction in batteries and PV technologies will drive the PV-battery configuration (Schmidt et al., 2017; Kittner et al., 2017; Ziegler and Trancik, 2021; Ziegler et al., 2021).

This study highlights the critical role of low-cost renewable electricity in achieving deep defossilisation of the energy systems via direct electrification of multiple end-use segments and indirect electricity use through power-to-X processes, as shown in Fig. 20. A new system architecture is observed, wherein renewable electricity emerges as the prime energy carrier and together with e-hydrogen by electrolysers that is further synthesised into other power-to-X products. **Renewable electricity-based hydrogen is foreseen to be an essential energy vector through the transition**, mainly in the production of e-fuels (Galimova et al., 2023a; Lopez et al., 2023b; Ball and Wietschel, 2009; Andrews and Shabani, 2012; Parra et al., 2019; Galimova et al., 2023c), e-chemicals (Galimova et al., 2023a; Lopez et al., 2023b; Dieterich et al., 2020; ElSayed et al., 2023), and e-materials (Lopez et al., 2023a; Keiner et al., 2023a; Otto et al., 2017; Palm et al., 2016; Lopez et al., 2022; Devlin and Yang, 2022; Bailera et al., 2021). The hydrogen requirement during the transition increases significantly from around 171 TWh in 2030 to about 4405 TWh in 2050 (see Fig. 20). The relevance of hydrogen is spurred by significant cost reduction in renewable electricity, especially of solar PV in Africa. Thus, the Power-to-X Economy in Africa could more appropriately be called a **Solar-to-X Economy**. It is clear that the complete substitution of hydrocarbons with renewable electricity is not technically feasible, as electricity cannot be used directly for some energy-industry demands such as long-haul flights and shipping, chemicals, and steel production (Lopez et al., 2023a; Galimova et al., 2023a). Therefore, renewable electricity-based synthetic fuels, chemicals, and materials are required to fulfil these demands through power-to-fuels, power-to-chemical, and power-to-materials. These core characteristics underscore the critical role of the Power-to-X Economy in building a sustainable energy system and effectively defossilising the energy-industry system.

It is noteworthy that power-to-X products are secondary energy carrier, as an indirect electrification pathway, they are subject to additional conversion losses during their production and utilisation phases (see Fig. 20). Thus, their renewable electricity requirements are higher than for direct electrification alternatives. However, in an integrated system, waste heat is recovered from electrolysis and synthesis of hydrocarbons, is used in the DAC units and industrial processes, thus improving the overall system efficiency. The relative efficiency gain of an energy system based on 100% RE across energy-industry system in

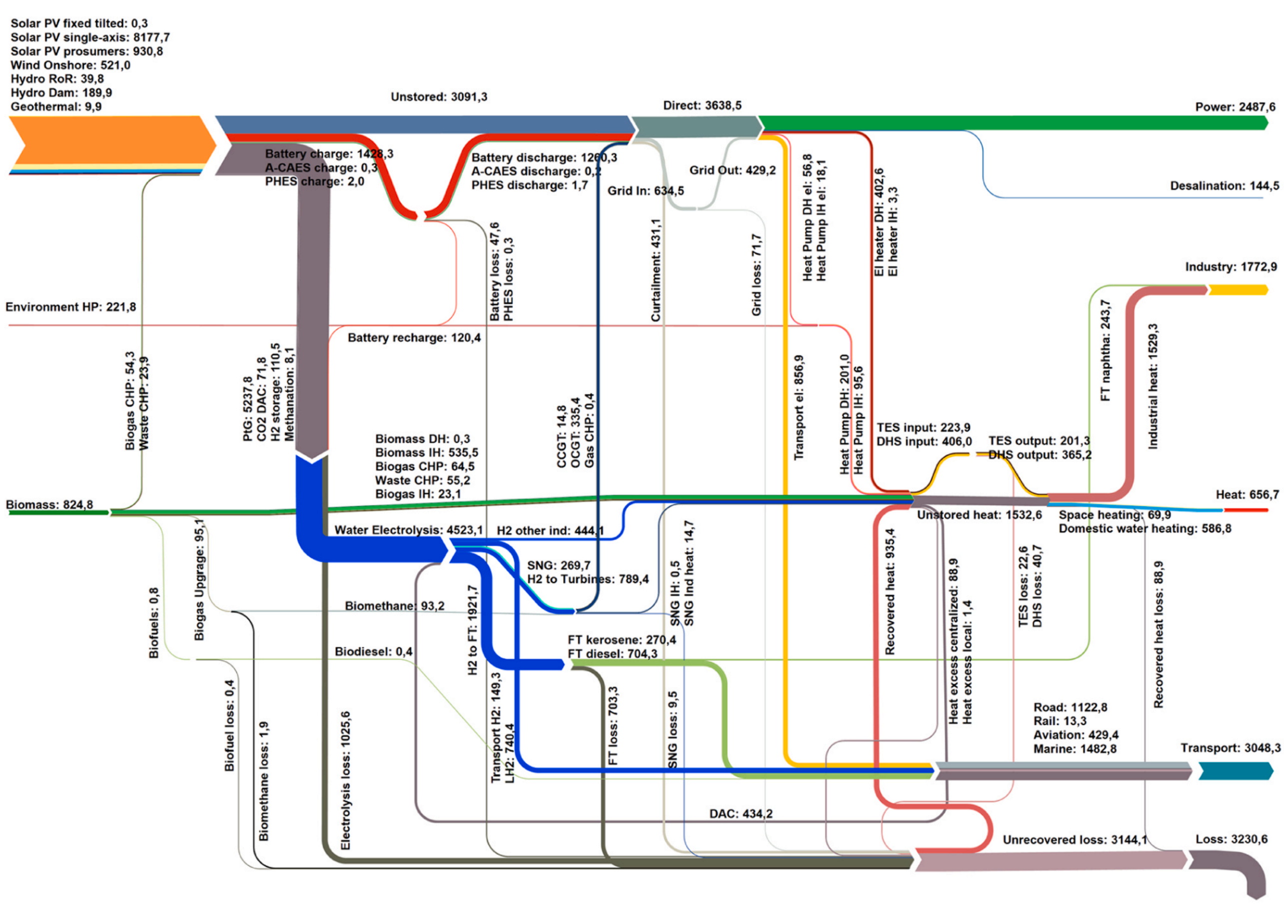

Fig. 20. Power-to-X Economy for the case of Africa for an energy-industry system with zero CO_2 emissions by 2050.

2050 compared to the efficiency and energy conversion standard as of today can be accounted to 57% (Oyewo et al., 2022), despite energy demand for e-fuels, as this energy demand is overcompensated by the enormous gains in direct electrification, as also found by others (Luderer et al., 2022; Eyre, 2021). Notably, direct electrification cost advantage increases with battery electric vehicles, heat pumps and steel production in electric arc furnace, mainly due to efficiency advantages.

4.2. Power-to-X Economy – fundamental conceptualisation of future energy-industry systems

The foundation of the African energy-industry system as a Power-to-X Economy and widespread direct and indirect electrification expands and improves upon the previously envisioned Hydrogen Economy. The original framing of the Hydrogen Economy in 1972 had been to provide hydrogen to the transport sector using electricity from nuclear power, thereby substituting fossil fuels, and limiting air pollution (Bockris, 1979). Within this framework, the historical 'hydrocarbon society' would transition to a 'hydrogen society' (Dunn, 2002). Additional early conceptualisations of the Hydrogen Economy have had their historical roots in solar PV (Bockris, 2002) and wind power (Armaroli and Balzani, 2011; Sherif et al., 2005). Although solar PV was identified as the core electricity source for hydrogen as early as 1999, the concept of a fossil fuel-based Hydrogen Economy has still lingered in the scientific discourse to this decade (Samsatli et al., 2011; Quarton et al., 2019; van der Spek et al., 2022). To encompass all energy and industry sectors, the Hydrogen Economy has been broadened to Hydrogen-to-X Economy frameworks, such as the Methanol Economy (Olah, 2005), which implies widespread hydrogen usage without direct handling and widespread usage of existing fossil fuel infrastructure.

A fundamental limitation of the Hydrogen Economy framework is that the direct substitution of hydrogen and hydrogen-based fuels typically ignores the impact of solar PV coupled with low-cost batteries and gains from direct electrification, which were found to be the major drivers of the African energy transition in this research. Therefore, the term 'Hydrogen Economy' may be misleading given the wide range of direct electric solutions (Ball and Weeda, 2015) and fails to incorporate the importance of low-cost renewable electricity as the major energy vector (Andrews and Shabani, 2012). The term 'Power-to-X Economy' (Ueckerdt et al., 2021; Galimova et al., 2023a; Palys and Daoutidis, 2022; Breyer et al., 2024), which has been widely used in this research, thus emerges as the more appropriate term for the widespread application of power-to-X routes for gases (e-hydrogen and e-methane), liquid e-fuels, chemicals (e-ammonia and e-methanol), heat, desalination, and materials (e-steel, e-carbon fibres, e-silicon carbide, etc) (Palys and Daoutidis, 2022; Sterner and Specht, 2021). Additionally, given the relevance of solar PV for the African energy system, the term 'Solar-to-X Economy' may be the more correct term for the African energy-industry transition.

4.3. Relevance of Power-to-X Economy for Africa – People-driven energy economy

The emergence of low-cost renewables, in particular solar PV, that is also abundantly available across Africa can trigger **new industrial opportunities and export business** for green e-hydrogen, e-fuels, e-chemicals, and e-materials (e-steel, e-plastic, e-carbon fibres, e-silicone carbide), which are increasingly being explored for North African countries, such as Morocco (Lopez et al., 2023a; Hampp et al., 2023; Hank et al., 2020; Galimova et al., 2023c), Egypt (Hampp et al., 2023; ElSayed et al., 2023), the Maghreb (Fasihi et al., 2017), and all of North Africa (van der Zwaan et al., 2021; Timmerberg and Kaltschmitt, 2019). Africa will benefit greatly from the expected restructuring of the global energy supply value chain. This, in turn, will boost employment at various stages of the value chains, including manufacturing, construction and installation, operation, and maintenance (Ram et al., 2022b). Job creation is one of the most important socio-economic footprints to measure the performance of the energy transition (Ram et al., 2022b; Bischof-Niemz and Creamer, 2019; Jacobson et al., 2019b). Scientific evidence has suggested a positive link between the RE transition and job creation. These studies have demystified the frequent asked question of whether the RE sector can realistically create the number of jobs as in the fossil-based system. In addition, there is a need to build **local expertise**, i.e., reskilling and training of personnel to enable switching from conventional fossil fuels to advanced renewable-based jobs. This, in turn, offers technologically advanced countries and regions the opportunity to export technical **know-how** to Africa and other countries and regions in the Global South with excellent renewable resources, where there is a unique opportunity to use low-cost renewable electricity to produce low-cost e-fuels, e-chemicals, and e-materials. As one of the most important potential global markets for power-to-X products, Africa has tremendous opportunities to leapfrog from its current state into a diversified and well-developed economy through income and investment in green e-fuels, e-chemicals, and e-materials production and value chains. A recent analysis for Egypt, Kenya, Mauritania, Morocco, Namibia, and South Africa, identifies commercial opportunities, job creation, export market and investment need of green hydrogen in these countries (Climate Champions, 2022). The study shows that by 2050 green e-hydrogen could sustainably industrialise these countries, increase their gross domestic product by 126 bUSD, which is equivalent to 6 – 12% of their current gross domestic power (IEEFA, 2022). Another study by the European Investment Bank (EIB), reports that Africa could make 1 trillion € (1.06 trillion USD) from green e-hydrogen a year by 2035, enabling it to export the fuel and boost local industry (IEEFA, 2022). Significant investment in capital expenditure will be required in power technologies, particularly solar PV, battery storage, electrolysers and other fuel-producing technologies in Africa.

4.4. Role of Africa in global defossilisation – Africa in an evolving global context

With its abundant RE resources, particularly solar energy, Africa is an optimal location for green e-hydrogen, i.e., hydrogen produced by electrolysis using its low-cost solar PV, as well as derived e-fuels, e-chemicals, and e-materials, which are critical components of the energy transition, and enable emissions reductions in hard-to-electrify sectors. Owing to their versatility, power-to-X hydrogen, fuels, feedstocks, and materials, can be stored, transported, and traded globally. Against this backdrop, **Africa can support the international efforts to mitigate climate change** by becoming a global market for e-fuels, e-chemicals, and e-materials, especially European countries seeking to diversity their energy import sources and meet their climate goals. However, it is worth noting that transportation costs of power-to-X products play an important role in influencing trade. For instance, hydrogen trade may not be attractive for long distances because of its prohibitively high transportation costs (Lopez et al., 2023a; Galimova et al., 2023c; Lopez et al., 2024; Schorn et al., 2021). However, there is an opportunity for African countries to export e-ammonia, e-methanol, e-FTL, e-LNG, e-steel products (e-HBI, e-CS), e-plastics (e.g., e-PE), and e-carbon fibres to Europe at low-cost, as our analyses show.

Trading of power-to-X products would enable the supply of low-cost e-fuels, e-chemicals, and e-materials to countries with inherent resource constraints, i.e., limited availability of land and potential for RE resources, to diversify supply as energy importers and accelerate their RE transition. This is the case for EU countries, which can benefit from high volumes of power-to-X product imports, as highlighted in this research. In this light, Africa and regions that have excellent RE sources year-round can **contribute to global energy security** and **create global partnership** for mutual benefit and business opportunities. The current Russian invasion of Ukraine has significantly disrupted economic activities globally, as energy and other commodity prices have been sky-rocketed (IEA, 2022). To wean off Russian gas in the EU, as rapidly and comprehensively as possible, Europe seeks to accelerate the green energy transition (Kemfert et al., 2022). In so doing, EU countries seek bilateral agreement on green e-fuel imports to support their domestic production. Against this backdrop, the EU-Africa partnership can be maximised for an excellent business environment for carbon-neutral export opportunities for African countries. Preliminary project plans in Namibia have already been discussed with a potential to produce 2 Mt of green ammonia by 2029 and potential customers in Europe and South Korea (Creamer Media's. Namibia signs agreement opening way for feasibility study of $10bn green ammonia project, 2023). Earlier proposals for partnerships include an interconnected grid between Europe and North Africa based on RE, called the Desertec vision (Zickfeld and Wieland, 2012; Breyer et al., 2020b). In sum, power-to-hydrogen, e-feedstocks, e-fuels, and e-materials emerge as the most effective options to accelerate the transition of hard-to-abate and increase the uptake of renewable electricity, through the indirect electrification approach. That may emerge as the real Desertec (Zickfeld and Wieland, 2012; Matthes et al., 2020).

4.5. Policy support: strategic actions are required to steer the Power-to-X Economy in Africa

Beyond the products and processes, policy measures are required to drive the Power-to-X Economy. Policy measures are required across the power-to-X value chains including production, infrastructure and end-use.

4.5.1. Charity begins at home: Africa energy policy should have power-to-X solutions at its core

Africa will emerge as one of the leading global players in power-to-X products made from renewables, as depicted in this research and others (Ueckerdt et al., 2021; van der Zwaan et al., 2021; Timmerberg and Kaltschmitt, 2019; Schorn et al., 2021; Matthes et al., 2020; van Wijk and Wouters, 2021). A number of hydrogen projects are underway or under discussion in South Africa, Namibia, Morocco, Mauritania, Egypt and Kenya (IEA, 2022). Research (Sterl et al., 2024; van Wijk and Wouters, 2021; Cardinale, 2023; Bhandari, 2022; Mukelabai et al., 2022; Restelli et al., 2023; Sadiq et al., 2023; Schöb et al., 2023), media (Climate Champions, 2022; IEEFA, 2022) and leading international reports (IEA, 2022; IRENA, 2022b) have identified the possible role of Africa in an evolving global context as a potential site for low-cost power-to-X products for international exports. Contrarily, the continent is characterised as poor in terms of access to modern energy services (IEA, 2022, 2019b; Casati et al., 2023; Sayani et al., 2021; Moner-Girona et al., 2021), while research on highly RE system analyses is scarce in Africa (Oyewo et al., 2023; Hansen et al., 2019). In this context, more research is required in Africa; more importantly, African energy policy should have direct electrification and indirect electrification based on renewables, well engrafted at its core. Such policies should focus on research and development programmes, developing vision documents, well-defined roadmaps and national strategies (IRENA, 2020b; Agora Energiewende and Guidehouse, 2021). African countries should launch a strategy for the production of power-to-X products, as well as compatible technologies, to boost their socio-economic development. Africa should not just be the production centre of power-to-X products for international export and yet remain poor as the case with the incumbent fossil crude oil exports (Adeola et al., 2021). Policy measures are required to maximise the potential of the Power-to-X Economy in Africa and avoid pitfalls.

4.5.2. Cost is the main driver of Power-to-X Economy: initiating policies to deliver low-cost power-to-X products

Given that the methodology is based on least-cost optimisation, this research finds Africa an excellent site for low-cost power-to-X products driven by high solar potential and low-cost solar PV. However, stable and supportive policy measures will be required to promote a sustainable Power-to-X Economy in Africa. This research outlined the crucial role of low-cost renewable electricity and hydrogen in the power-to-X value chain. More importantly, low-cost renewable electricity is critical to the overall cost competitiveness of power-to-X products. In this context, policies are required to support that electricity is sustainable and available at a low cost. Notably, WACC rates are a decisive and substantial factor in determining the profitability of estimates (Vartiainen et al., 2020; IRENA, 2022b; Egli et al., 2019, 2018; Lonergan et al., 2023; Ondraczek et al., 2015); countries with excellent resource conditions and low WACC would emerge as the leading exporters of power-to-X products (IRENA, 2022c). Currently, WACC and capex in many African countries are still high (Egli et al., 2019). However, it is expected to decline with economic development (Vartiainen et al., 2020; Kutscher et al., 2010). Power -to-X products are unlikely to be cost-competitive in the near future without technological development and scale-up (Böhm et al., 2020), which is required to reduce the unit cost of major components, particularly solar PV, electrolysers and other fuel-producing technologies. This study highlights the critical role of solar PV as the primary source of low-cost electricity in Africa, and the capex of solar PV is expected to decline and converge over time.

In summary, **cost is a deciding factor**. Using energy system models to support clean energy spending allocations is often helpful. These models help policymakers to identify optimal investment paths by illustrating, among other things, the system design and costs that result from the policies under consideration. The dynamics for international trade of power-to-X products will significantly be affected by cost differences worldwide depending on macroeconomic stability, political uncertainties and maturity of financial markets in different countries (Egli et al., 2019; Ondraczek et al., 2015). Against this backdrop, policies are required in emerging and developing countries to make low-cost finance available through de-risking measures (Waissbein et al., 2013; Schmidt, 2014). In addition, policy measures are required to cater to barriers related to technology, cost gap, market development and regulation, standardisation, financing, infrastructure, pace and scale of deployment (IRENA, 2020b, 2022c; Velazquez Abad and Dodds, 2020).

5. Limitations, concerns, and opportunities for future research

Further research is required in Africa, as most studies have predominantly a focus on North Africa while many other countries in the region with excellent renewable resources are yet to be researched on the topic. Such findings will be beneficial for Europe seeking to diversify its energy supply imports while further importing regions would benefit as well foremost Northeast Asia (Galimova et al., 2023a).

The research into e-steel and e-plastics has several limitations that can be considered in future research. For the e-steel supply chains, electrowinning, which would allow for a nearly complete electrification of steelmaking, is not considered; however, with technological progress and low-cost renewable electricity supplies, this process could be competitive with H-DRI-EAF steelmaking (Lopez et al., 2022; Bailera et al., 2021). Further research could similarly be conducted investigating the global LCOCS for green steelmaking to identify the least cost HBI and CS production sites. Research on e-carbon fibres is required to better understand the substitution potential for e-steel (Keiner et al., 2023a). For the e-PE supply chain, the focus was placed on e-methanol, though there are additional routes that can be utilised to produce e-ethylene e.g., e-methane via oxidative coupling of methane (Zhao et al., 2021). Similar supply chain configurations and those considered in this study could be applied to those routes to identify the least cost supply chain configurations among all potential power-to- X routes, especially considering the importance of feedstock costs on cost-competitive e-plastics. e-Ethylene exports may also be considered, as ethylene can be traded via pipelines, which may be relevant for short pipelines between the countries of North Africa and Europe.

The trading model used to identify potential exporters and volumes of traded e-fuels and e-chemicals has several limitations. This study assumed fixed shipping costs of 1.5–5 €/MWh, depending on the type of e-fuel and e-chemical. However, the transportation costs, and consequently the trading results, could change depending on the distance, capacities of each of the components of the transportation chain, and means of transport. It is likely that the shipping costs are underestimated, in particular for ammonia, for which shipping costs have been estimated to cost 7.2–14.1 €/MWh for a distance of 13,000–14,600 km (Galimova et al., 2023b). In addition, existing infrastructure, in particular pipelines, or potential refurbishment of the infrastructure has not been considered. The current trading results showed almost no exports from North Africa. However, if existing pipeline network between North Africa and Europe would be considered, the trading results may have differed.

Lastly, the levelised costs of e-fuels and e-chemicals for Egypt have the following limitations: 1) they are calculated based on regional average LCOE, not high-resolution site-dependent costs. 2) the transportation costs to the destination country(ies) are not included. 3) In the early years (2025–2030), some fossil fuels were used to produce these e-fuels due to the imposed limit on RE growth. 4) A share of 10% of the European demand for e-fuels and e-chemicals was assumed for exports from Egypt, and no sensitivity analysis was carried out to determine the optimum level of exports. Further research will be required to address these limitations. The analysis shows that CSP-TES and the PV-battery-HP system configurations are comparable and technically feasible solutions. However, a more detailed analysis of each configuration's system flexibility and cost in higher spatial resolution and least cost site-based optimisation would help attain a more conclusive result. It is worth noting that only CO_2 DAC was the only option investigated for the case of Egypt's e-fuel and e-chemical production. More detailed research is required considering resource availability, societal preference, and the techno-economic impact of the CDR portfolio on the energy system. In addition to the mentioned limitations, the way how e-fuels and e-chemicals modelling is done within energy system models requires revision, since more and more new greenfield projects are partly or fully off-grid e-fuel systems, as this may be the least cost case, while a partial grid connection for electricity and hydrogen may be the overall system optimum.

6. Conclusions and policy implications

This research finds that **Africa is a hotspot for green e-fuels, e-chemicals, and e-materials supply to Europe**. Against this backdrop, four conclusions are offered for energy analysts, planners, and policymakers.

First and positively, **Africa is an important location for low-cost sites for green e-fuels, e-chemicals, and e-materials production** based on its abundant renewable energy, particularly solar energy. The term Power-to-X Economy in Africa should be Solar-to-X Economy, as solar PV emerges as the prime source of energy across the continent due to its excellent conditions and an anticipated decline in the cost of technology. The cost-optimal pathway for Africa favours PV-battery hybrid system configuration. The PV-battery hybrid systems will enable an accelerated transition, rapid electrification, and decentralised variable renewable energy ramping.

Second, **Africa is well positioned as an exporter of green e-fuels, e-chemicals, and e-materials**, backed by its cheap and excellent renewable energy resources. Owing to their properties, renewable electricity-based hydrogen, fuels, chemicals, and materials are storable and transportable. However, the deciding factor is their cost of production. The results show that Africa's low-cost power-to-X products backed by low-cost renewable electricity, mainly supplied by solar PV, is the basis for Africa's vibrant export business. Power-to-X products are fully tradable from Africa, depending on transport costs. Our analysis shows that hydrogen will likely not be traded simply due to high transport costs. However, there is an opportunity for African countries to export e-ammonia, e-methanol, e-FTL, e-LNG, e-steel products, e-plastics, and e-carbon fibres to Europe at low cost.

Third, **Africa can be sustainably industrialised, support a global defossilisation strategy and achieve a diversified and well-developed economy by producing green e-hydrogen, e-fuels, e-chemicals, and e-materials** with its low-cost renewable electricity. With its cheap and excellent renewable resources, Africa can support the global defossilisation strategy, particularly in European countries seeking to diversify their energy import sources while fulfilling their climate targets. The ongoing Russian war in Ukraine has intensified the shift towards a sustainable energy transition. This is the case for many countries worldwide, especially the European countries, which are revamping their energy policy to fast-track the green energy transition. However, due to resource limitations, these countries would need to import green e-fuels, e-chemicals, and e-materials, in addition to their local production, to meet their demand. The power-to-X approach will dominate the energy-industry system through direct or indirect electrification, which will enable deep defossilisation of the hard-to-abate demand, including long-distance shipping and aviation, industrial feedstocks, and material refining. With its rich-renewable resources, Africa will create new industrial opportunities and jobs and attract significant investment in a Power-to-X Economy. The emergence of new green e-fuels, e-chemicals, and e-materials industry presents a great opportunity for Africa to leapfrog from its current state into a diversified and well-developed economy. Massive investment will be required in power generation technologies, particularly solar PV, battery storage, electrolysers and other fuel-producing technologies in Africa.

Four, **right policy environment and continued research** on the topic will be required in Africa. Stable and supportive policy measures are required to create an enabling environment for Africa to benefit from the energy transition. Africa's energy policy should have renewable electricity-based e-fuel, e-chemicals, and e-materials production at its core for domestic consumption and international exports. Policy addressing standardisation, multilateral exchanges and barriers to investment will be required. Further research on the topic is required in Africa, as most analyses have predominantly focused on North Africa while less research is given to other countries in the region. More importantly, such analyses will be required for countries along the coastline, especially in the Horn of Africa, with excellent solar and wind energy resources. In sum, market players, policymakers and research experts will need to collaborate and coordinate future activities to deploy and scale-up investments in power-to-X solutions.

CRediT authorship contribution statement

Gabriel Lopez: Writing – review & editing, Visualization, Formal analysis, Data curation. **Mai ElSayed:** Visualization, Formal analysis, Data curation. **Tansu Galimova:** Writing – review & editing, Visualization, Investigation, Formal analysis, Data curation. **Christian Breyer:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Ayobami Solomon Oyewo:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

None

Data Availability

No data was used for the research described in the article.

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